Supplementary Online Content

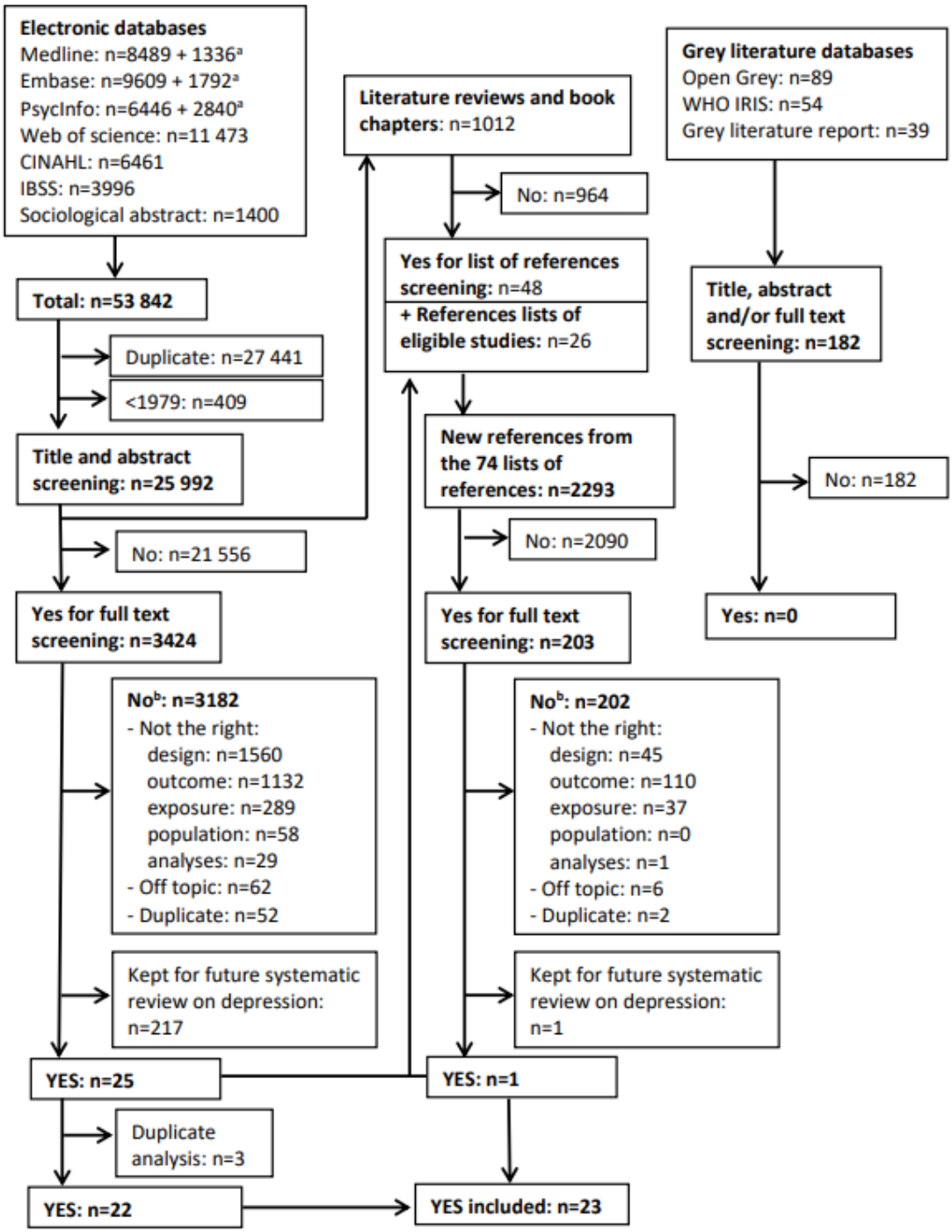
Duchaine CS, Aubé K, Gilbert-Ouimet M, et al. Psychosocial stressors at work and the risk of sickness absence due to a diagnosed mental disorder: a systematic review and meta-analysis. *JAMA Psychiatry.* Published online April 1, 2020. doi:10.1001/jamapsychiatry.2020.0322

eFigure 1. Detailed Flowchart for the Selection of the Included Studies
eFigure 2. Funnel Plot of Studies on the Effect of A) Job Strain, B) High Psychological Demands and C) Low Social Support at Work on the Risk of Sickness Absence Due to a Diagnosed Mental Disorder
eTable 1. Characteristics and Results of the 23 Included Studies
eTable 2. Results of the Sensitivity Analyses
eAppendix 1. Search Strategy for Each Electronic Database, First Run in October 2017 and Updated for Medline, Embase and PsycInfo in Febuary 2019
eAppendix 2. Completed ROBINS-I Tool Assessment Grid
eAppendix 3. Quality Evaluation of Prospective Studies on Psychosocial Stressors at Work and Sickness Absence Due to a Diagnosed Mental Disorder, According to Risk of Bias in Non-randomised Studies-Intervention Tool (ROBINS-I) Criteria
eReferences.

This supplementary material has been provided by the authors to give readers additional information about their work.

© 2020 American Medical Association. All rights reserved.

eFigure 1. Detailed Flowchart for the Selection of the Included Studies



[a] From the February 2019 update

[b] Definition of reasons for exclusion: Not the right 1) study design: cross sectional, ecological, intervention studies, case-control with a retrospective design, qualitative or case study, and not an original study; 2) outcome: all-cause absence, self-reported absence or reason for absence, mental health not evaluated, non-validated measure of mental health or off topic; 3) exposure: psychosocial stressors at work from at least one of the three considered models were not measured or were measured with a non-validated tool; 4) population: population of patients or of workers on sick leave, pregnant women, in a process of return to work; 5) analyses: association between psychosocial stressors at work and mental health not reported (not an objective of the study).

© 2020 American Medical Association. All rights reserved.

eTable 1. Characteristics and Results of the 23 Included Studies

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
Roelen (2018) Norwegian SUrvey of Shift work, Sleep and Health (SUSSH) Norway [1]	Baseline: 2008, FU: 2 y, 2009-2010 Nurses working at least 50% of a full-time position, random sample 1533 / 5400 PB: 38%, MD: 26% 1381 / 145 (7 missing) Mean age: 33.1 y	Exposure at baseline JCQ [2] PD: 5 items, score range 5-20, mean: 14.3 SS: 6 items, score range 6-24, mean: 17.4 JC: 6 items, score range 6-24, mean: 17.7, Excluded from analyses due to its low Cronbach's α (0.52)	Mental health-related long-term sickness absence ≥17days, Statistics Norway records International Classification of primary care (ICPC) by general practitioner category P which corresponds to ICD-10 chapter-V (mental and behavioral disorders)	Cox proportional hazards model HR (95% CI) PCE: Unclear Age, sex, marital status, care for children at home 103 / 1533 PD: 1.05 (0.97-1.13) **SS: 0.92 (0.87-0.97)**
Vendrig (2018) NA Netherlands[3]	Baseline: NA FU: 6 months Employees of a home care organization participating in a health survey 388 / NA PB: NA, MD: NA 287 / 101 Mean age: 42.2 y SES: NA	Exposure at baseline Well-Being Inventory (WBI), validated with JCQ[3] T-scores in continuous, mean: NA PD: 8 items, named job strain in this article JC: 5 items SS: 5 items	Absenteeism because of common mental disorder ≥6 weeks, diagnosis established by an occupational physician Codes: NA	Logistic regression, OR (95% CI) PCE: Yes Crude 7 / 388 PD: NA (0.93-1.14) JC: NA (0.83 -1.01) SS: NA (0.89-1.07)
van Hoffen (2018) NA Netherlands [4]	Baseline: 2010 FU: 2 y, Jan 2011 to Dec 2012 Workers in the distribution and transport sector who participated in a health survey in 2010 2782 / 4877 PB: 82%, MD: 31% 1547 / 1235 Mean age: 49.9 y	Exposure at baseline Questionnaire on the Experience and Evaluation of Work [5], Score in continuous expressed as % of the maximum scale score, range 0-100 Workload: 11 item, mean: 44.9 Work pace: 7 items, mean: 39.6 Variety of work: 6 items,	Mental long-term sickness absence ≥42 days, obtained from an occupational health register, medically certified by an occupational physician ICD-10 (F codes)	Logistic regression OR (95% CI) PCE: Yes Age, gender, education 73 / 2782 Workload: 1.03 (0.85-1.25) Work pace: 0.96 (0.81-1.15) Variety in work: 0.96 (0.82-1.12) Autonomy: 0.91 (0.70-1.05)

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
	Education: low: 52%; middle: 37%, high: 11%	mean: 32.0 Autonomy: 3 items, mean: 33.2 Participation in decision: 6 items, mean: 24.4 Learning opportunities: 4 items, mean: 15.3 SS supervisor: 5 items, mean: 48.1 SS co-workers: 3 items, mean: 53.5		Participation in decision: 0.88 (0.72-1.07) Learning opportunities: 0.83 (0.65-1.05) SS supervisor: 0.93 (0.81-1.07) SS co-workers: 0.97 (0.85-1.11)
Ndjaboué (2017) Quebec white-collar workers, Canada [6]	Baseline: could be at T0 (2000-2003), T1 (after 3 y) or T2 (after 5 y) FU: mean: 4.88 y White-collar workers from three public insurance organizations 2082 / 2515 PB:81%, FU: 86-85%, MD: 2% 1319 / 763 Men: <40 y: 29.1%; 40-49 y: 43.1%; ≥50 y: 27.8%; Women: <40 y: 35.9%; 40-49 y: 47.7%; ≥50 y: 16.5%; Education: Men: secondary or less: 12.2%; College: 32.2%; University: 55.6%; Women: secondary or less: 30.2%; college: 33.8%; university: 36%	Exposure at baseline JCQ French version [2,7] PD and JC: 9 items each, dichotomized using median split from external sample JS: quadrant: High strain: Men: 18.87% Women: 25.78% Adapted from the French version of Siegrist's questionnaire [8], 15 items ERI: (E/R ratio, ≤1 or >1) ERI >1: Men: 27.3%; Women: 28.4% Combined exposure of JS and ERI: 4 categories	Medically certified absence for mental disorders of ≥5 days Employer registers and phone interview with workers for unavailable diagnosis (5.3% of episodes of absence) ICD-10: F10-F19, F30-F39, F40-F48, F50-F59, F60-F69, F99, Z56.7, G47.9, Z65.8, R53	Andersen-Gill Cox regression model, HR (95% CI) PCE: Yes Age education, marital status, stressful events, homeload, smoking, BMI, sedentary behavior, alcohol intake All: 281 / 2085, Men: 59 / 763 JS: 2 categories: High strain: 0.71 (0.38-1.30) Quadrant: High strain: 0.68 (0.31-1.49) Active: 0.96 (0.47-1.95) Passive: 0.93 (0.47-1.85) Women: 222 / 1319 JS: 2 categories: **High strain: 1.69 (1.29-2.21)** Quadrant: **High strain: 2.55 (1.55-4.20)** Active: 1.65 (0.98-2.79) Passive: 1.59 (0.97-2.60) ERI: See Ndjaboué (2014)

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
Ohta (2017) NA Japan [9]	Baseline: 2012 FU: 1 y, 2013 Workers from an information technology company in Tokyo 1408 / 1667 PB: 93.5%, FU: 84%, MD: NA 247 / 1161 Mean age: 36.8 y Job position: Management: 21%	Exposure at baseline The Brief Job Stress Questionnaire (BJSQ) [10,11] JS: ratio: PD: 3 items / JC: 3 items, mean: 1.1 SS: 6 items, sum score, range: NA, mean: 16.0	At least one sickness absence for mental illness >7 days obtained from employees' computerized records, diagnosed by occupational physician Codes: NA	Logistic regression OR (95% CI) PCE: Unclear Age, sex, job position, change in the WAI score (subjective estimation of work ability, number of diagnosed diseases, subjective estimation for work impairment due to diseases, sickness absences during the past year, mental resources), job support, job stress 17 / 1408 JS: 0.44 (0.16-1.00) SS: 0.88 (0.75-1.02)
Juvani (2016) Finnish Public Sector Study (FPS) Finland [12]	Baseline: 2000-2002 and 2004 FU: Jan 2005 to Dec 2011 Mean: 6.4 y Blue- and white-collar workers of 10 towns and 6 hospitals 24 895 / 71 705 PB: 68%, FU: 80%, MD: 13% 20 271 / 4624 Age: <40 y: 16,6%; 40-50 y: 36,2%; 50-60 y: 42,5%; ≥60 y: 4,7% Education: elementary: 8.5%; secondary: 34.2% tertiary: 57.3%	Exposure combination from 2 time points (2000-2002 and 2004) Moorman's (1991) scales OJ: mean of 2 time points continuous, 0-5, mean: 3.31 RJ: 6 items, mean of 2 time points continuous, 0-5, mean: 3.65 PJ: 7 items, mean of 2 time points, continuous, 0-5, mean: 3.01	Disability pension due to depression, >300 days Government registers ICD-10 F32-F34	Cox proportional hazards model HR (95% CI) PCE: In sensitivity analyses only, but not for all models or all psychosocial stressors at work Age, sex, location of workplace, occupational status, education and size of residence 1-All: 282 / 24 895 Men: 28 / 4624 Women: 254 / 20 271 Ref=low **OJ: All: 0.69 (0.58-0.82)** **Men: 0.49 (0.30-0.81)** **Women: 0.71 (0.59-0.86)**

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
				RJ: All: 0.75 (0.65-0.86) **Men: 0.53 (0.35-0.80)** **Women: 0.77 (0.67-0.90)** **PJ: All: 0.76 (0.65-0.89)** **Men: 0.62 (0.39-1.00)** **Women: 0.78 (0.65-0.92)**
Mather (2015) Study of Twin Adults: Genes and Environment (STAGE) Sweden [13]	Baseline: Nov 2004-May 2006 FU: Baseline to Dec 2010; ≈5 y Twins born between 1959 and 1985 from the population-based Swedish Twin Registry 11 729 / 42 582 PB: 59.9%, FU: NA, MD: 36% 5766 / 5963 Mean age: 35 y, range: 20-47 y Education: elementary: 5%; secondary: 48%; higher: 47%	Exposure at baseline JCQ Swedish version [14] JS: quadrant, median split, high strain: 31% PD: 5 items, continuous, 1-4, mean: 2.5 JC: 5 items, continuous, 1-4, mean: 1.9 SS: 6 items, continuous, 1-4, mean: 1.6 1=best; 4=worst IS: High strain + low social support: 17%	Sick leave spells due to mental disorder of ≥7 days Government registers ICD-10: F00-F99	Logistic regression OR (95% CI) PCE: Yes Age, sex, education, previous sick leave, self-rated health 972 / 11 729 JS: Passive: 1.02 (0.81-1.28) **Active: 1.30 (1.02-1.67)** High strain: 1.40 (1.12-1.75) **PD: 1.41 (1.24-1.60)** JC: 1.02 (0.90-1.17) SS: 1.15 (0.98-1.36) **IS: 1.47 (1.16-1.88)**
Janssens (2014) Belstress III Belgium [15]	Baseline: 2004-2005 FU: 1y after the completion of the questionnaire Workers of seven companies or public administrations 2640 / ≈ 9813 PB: 30.4%, FU: 96%, MD: 8% 1414 / 1226 Mean age: 43.3 y, range: 30-55 y Education: ≤3 y of secondary: 20.8%; Secondary complete: 34.7%;	Exposure at baseline JCQ French version [16] JS: PD/JC ratio, continuous, 0.13-1.58, mean: 0.46 PD: continuous, 5 items, 12-48, mean: 30.4 JC: continuous, 11 items, 24-96, mean: 68.2 SS: continuous, 8 items, 8-32, mean: 22.8 Siegrist's questionnaire, French version	Long-term sickness absence due to mental disorders ≥15 days Employer registers and General practitioner of the worker for the cause Codes: NA	Logistic regression OR (95% CI) PCE: Unclear Age, alcohol consumption, baseline depressive symptoms JS: 1.06 (0.85-1.33) PD: 0.90 (0.71-1.15) JC (ref=low): 0.81 (0.65-1.02) SS (ref=low): 0.83 (0.65-1.05) E: 0.91 (0.71-1.17) **R (ref=low): 0.76 (0.60-0.97)** ERI: 1.11 (0.88-1.39)

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
	College complete: 44.5%	ERI: E/R ratio continuous, 0.28-3.67, mean: 1.1 E: 5 items, continuous, 5-20, mean: 14.2 R: 11 items, continuous, 11-44, mean: 29.4		
Juvani (2014) Finnish Public Sector Study (FPS) Finland [17]	Baseline: 2000-2002 FU: to Dec 2010, mean: 8.9 y Blue- and white-collar workers of 10 towns and 6 hospitals 1-Work-unit level exposure: 51 874 / 71 705 PB: 100%, FU: 99%, MD: 14% 39 089 / 12 785 Mean age: 44.3 y, range: 17-64 y Education: Primary: 13.7%; Secondary: 35.8% Tertiary: 50.5% 2-Individual level exposure: 35 260 / 71 705 PB: 68%, FU: 99%, MD: 42% W / M, age and SES: NA	Exposure at baseline Adapted from Siegrist's questionnaire ERI: E/R ratio in quartiles, Lowest (Q1): Work-unit level: 25% Individual level: 27% E: 1 item, quartiles R: 3 items, quartiles Mean or prevalence: NA	Disability pension due to depression, >300 days Government registers ICD-10 F32-F34	Cox proportional hazards model HR (95% CI) PCE: Yes Age, sex, place of residence, occupational status, education, income, baseline physical illness, baseline mental disorder, type of employer, type of work contract, size of work unit, mean age of employees in work unit, proportion of fixed term workers in work unit and work unit level of job strain 1-Work-unit level: 890 / 51 874 ERI: Q2: 1.21 (0.98-1.49) Q3: 1.20 (0.97-1.49) **Q4: 1.63 (1.31-2.04)** E and R: SES not included in the model E: Q2: 1.02 (0.84-1.23) Q3: 0.90 (0.74-1.10) Q4: 0.99 (0.82-1.20) **R: Q1: 1.72 (1.42-2.08)** **Q2: 1.28 (1.05-1.57)**

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country, (reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
				Q3: 1.14 (0.93-1.39) 2-Individual level: 551 / 35 260 ERI: Q2: 0.97 (0.73-1.29) Q3: 1.18 (0.92-1.51) **Q4: 1.90 (1.51-2.40)** E and R: SES not included in the model E: Q2: 0.64 (0.36-1.16) Q3: 0.64 (0.37-1.10) Q4: 0.83 (0.48-1.43) **R: Q1: 1.92 (1.49-2.48)** **Q2: 1.50 (1.11-2.01)** Q3: 1.11 (0.84-1.45)
Ndjaboué (2014) Quebec white-collar workers, Canada [18]	Baseline: could be at T0: (2000-2003), T1 (after 3 y) or T2 (after 5 y) FU: mean: 4.88 y White-collar workers from three public insurance organizations 2086 / 3987 PB: 80.9%, FU: 86-85%, MD: 5% 1321 / 765 Age range: 41-56 y Men: <40 y: 29%; 40-49 y: 43%; ≥50 y: 28%; Women: <40 y: 35.8%; 40-49 y: 47.6%; ≥50 y: 16.6%;	Exposure at baseline Adapted from the French version of Siegrist's questionnaire, 13 items, [8] ERI: E/R ratio, ≤1 or >1 ERI >1: Men: 27.3%; Women: 28.3% E: 4 items, tertiles Men: High: 40.5%; Women: High: 36.3% R: 11 items, tertiles Men: low: 36.7%; Women: low: 32.3%;	Medically certified absence for mental disorders of ≥5 days Employer registers and phone interview with workers for unavailable diagnosis (5.34% of episodes of absence) ICD-10: F10-F19, F30-F39, F40-F48, F50-F59, F60-F69, F99, Z56.7, G47.9, Z65.8, R53	Cox proportional hazard model HR (95% CI) PCE: Yes Age, sex, education, All: 281 / 2085 Men: 59 / 765 Women: 222 / 1321 **ERI: All: 1.42 (1.11-1.85)** Men: 1.20 (0.69-2.09) **Women: 1.47 (1.11-1.94)** E: (ref=low) , All: med: 1.13 (0.83-1.54); high: 1.10 (0.81-1.50); Men: med: 1.12 (0.57-2.19); high: 0.89 (0.45-1.74); Women: med: 1.15 (0.81-1.67); high: 1.18 (0.83-1.67); R: All: med: 1.29 (0.95-1.76); **low: 1.57 (1.17-2.10);**

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
	Education: Men: secondary or less: 12.16%; College: 32.3%; University: 55.6%; Women: Secondary or less: 30.2%; College: 33.9%; University: 35.9%			**Men: med: 2.30 (1.07-4.97); low: 3.42 (1.66-7.06);** Women: med: 1.13 (0.80-1.60); low: 1.30 (0.94-1.79)
Elovainio (2013) Finnish Public Sector Study (FPS) Finland [19]	Baseline: 2000-2002 and 2004 FU: 1y Jan 2005 to Dec 2005 Blue- and white-collar workers from 10 towns and 21 public hospitals 17 641 / 71 705 PB: 68%, FU: 60%, MD: 9% 17 818 / 3403 Mean age: 47.7 y Occupation: Manual worker: 15%; Lower grade non-manual worker: 57%; Upper-grade non-manual worker: 28%;	Exposure combined from 2 time points (2000-2002 and 2004) Moorman's (1991) scales 1-Self reported RJ: 6 items, mean of 2 time points, continuous, 0-5, mean: 3.65 PJ: 7 items, mean of 2 time points, continuous, 0-5, mean: 3.03 2-Agregated from co-workers response RJ: 6 items, mean of 2 time points, continuous, 0-5, mean: 3.66 PJ: 7 items, mean of 2 time points, continuous, 0-5, mean: 3.01	1-Long-term sickness absence due to a diagnosed mental disorder, ≥9 days Government registers ICD-10 codes: F00-F99 855 cases 2-Long-term sickness absence due to diagnosed depression, ≥9 days Government registers ICD-10 codes: F32-F34	Logistic regression, OR (95% CI) PCE: Yes Age, sex, type of employment contract, SSE (five-digit occupational titles), length of employment, size of work place 1-Self-reported: 396 / 17 641 All mental health diagnosis: **RJ: 0.76 (0.67-0.86)** **PJ: 0.79 (0.69-0.92)** Depression only: **RJ: 0.78 (0.64-0.96)** **PJ: 0.75 (0.58-0.97)** 2-Aggregated from co-workers: All mental health diagnoses: 396 / 17 641 RJ: 0.84 (0.63-1.10) PJ: 0.83 (0.61-1.13) Depression only: RJ: 1.29 (0.78-2.13) PJ: 1.34 (0.77-2.33)
Samuelsson (2013) Swedish Twin Study of Disability	Baseline: 1993 FU: 12 y, to Dec 2008	Exposure at baseline Job exposure matrix based on exposure data from a	Disability pension due to mental disorder diagnosis Minimal days: NA	Cox proportional hazards model HR (95% CI)

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
Pension and Sickness Absence (STODS) Sweden [20]	Twins born between 1925 and 1958 from the population-based Swedish Twin Registry 42 715 / 59 893 PB: 100%, FU: 100%, MD: 1% 21 117 / 21 598 Mean age in 1990: 45 y, range: 32-62 y Education: DP with mental diagnosis: high: 28%, intermediate: 42%, low: 30%; DP other diagnoses: high: 13%, intermediate: 42%, low: 45%; No DP: high: 25%, intermediate: 42%, low: 33%;	population-based survey of self-reported PD, JC and SS assigned to individual in the study population according to occupational codes. [21] PD, JC and SS: continuous, 0-10 Means: DP mental diagnosis: PD: 6.05; JC: 6.42; SS: 6.48 DP other diagnoses: PD: 6.27; JC: 6.19; SS: 6.40 No DP: PD: 6.00; JC: 6.66; SS: 6.33 JS: quadrant, high strain: DP mental diagnosis: 12% DP other diagnosis: 14% No DP: 14% IS: high strain + low SS DP mental diagnosis: 9% DP other diagnoses: 9% No DP: 7%	Government registers ICD-10: F00-F99	PCE: Yes Age, sex, education, marital status, N children living at home and type of living area 1420 / 42 715 **JS: Passive: 1.26 (1.05-1.51)** Active: 0.90 (0.74-1.08) High strain: 0.89 (0.70-1.14) **PD: 1.11 (1.02-1.21)** **JC: 0.91 (0.87-0.96)** **SS: 1.12 (1.01-1.24)** **IS: 1.31 (1.04-1.65)**
Lahelma (2012) Helsinki Health Study Finland [22]	Baseline: 2000, 2001, 2002 FU: until end of 2008 Blue- and white-collar workers of Helsinki 6525 / 13 344 PB: 67%, FU: 74%, MD: 0% 5122 / 1403 Aged 40, 45, 50, 55 or 60 y at baseline Occupational class:	Exposure at baseline Based on JCQ PD: 9 items, dichotomized to upper quartile, High PD: Men: 20%, Women: 23% JC: 9 items, dichotomized to upper quartile, Low JC: Men: 24%, Women: 21% SS from Sarason's SS: 4 items, dichotomized to upper quartile, Low SS:	Disability retirement due to mental disorder Minimal days: NA Government registers ICD-10: F00-F99	Cox proportional hazards model HR (95% CI) PCE: Yes Age as time scale, sex, shift work, temporary work contract, working overtime, hazardous exposure, physical workload, computer work, job control, psychological demand,social

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
	Women: managers and professionals: 29%, semi-professionals: 19%, routine non-manual workers: 42%, manual workers: 11%; Men: managers and professionals: 45%, semi-professionals: 20%, routine non-manual workers: 9%, manual workers: 26%	Men: 32%, Women: 34%		support at work, occupational class 143 / 6525 **PD: 1.48 (1.00-2.18)** **JC: 1.67 (1.12-2.49)** SS: 0.82 (0.56-1.19)
Mantyniemi (2012) Finnish Public Sector Study (FPS) Finland [23]	Baseline: 2000-2002 FU: From Jan following the survey to Dec 2005 mean: 4.6 y Blue- and white-collar workers of 10 towns and 21 hospitals **1-Work unit based:** 62 805 / 71 705 PB: 100%, FU: 100% MD: 2% 48 012 / 14 793 **2-Occupational based**: 67 953 / 71 705 PB: 100%, FU: 100% MD: 2% 51 780 / 16 173 mean: 44.3 y; range: 17-64 y Occupation: Upper non-manual: 30.2%; Lower non-manual: 49.3%; Manual: 20.5%; **3-Individual based:** 45 579 / 71 705 PB: 68%, FU: NA, MD: 6% W / M and SSE: NA	Exposure at baseline Based on JCQ: Self-reported for a subset of the study population and Job exposure matrix based on responses of this subset PD: 3 items, JC: 9 items; JS: mean of PD minus mean of JC 1-Work unit-based Mean: 3.58 2-Occupation title-based: Mean: 3.57 3-Individual-based, mean: NA	Disability Pension due to depression, >300 days Government registers ICD-10: F32-F34	Cox proportional hazards model HR (95% CI) PCE: Yes Age sex, type of job contract, stratified by SES (occupation) 1-Work unit-based: 461 / 62 785 2-Occupation based: 493 / 67 953 3-Individual-based: 294 / 45 571 JS: 1-Work unit-based: **Upper non manual:** **1.56 (1.09-2.21)** Lower non-manual: 1.00 (0.80-1.26) Manual: 1.29 (0.93-1.79) 2-Occupation-based: Upper non manual: 1.67 (0.92-3.05) Lower non-manual:

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
				1.12 (0.83-1.51) Manual: 1.18 (0.76-1.81) 3-Individual-based: **Upper non manual:** **1.70 (1.36-2.13)** **Lower non-manual:** **1.29 (1.11-1.50)** **Manual: 1.49 (1.23-1.80)**
Inoue (2010) Japan Work Stress and Health Cohort (JSTRESS) Study Japan [24]	Baseline: 1996-1998 FU: from baseline to 1999-2003, mean: 5.14 y Factory workers 15 256 / ≈ 26 788 PB: 85%, FU: NA, MD: 19% Men only, Mean age: 40.5 y; 45 et + y: 35.7%; 35-44 y: 40.3%; 18-34 y: 24% Education: >12 y: 37.5%; ≤12 y: 62.5%	Exposure at baseline, National Institute for Occupational Safety and Health Generic Job Stress Questionnaire (NIOSH-GJSQ), Japanese version [25] JS: ratio in tertiles, mean: 0.85 PD: 11 items, tertiles, 11-55 mean: 37.4 JC: 16 items, tertiles, 6-42 mean: 47.5 Supervisor SS: 4 items, tertiles, 4-20; mean: 14.8 Co-worker SS: 4 items, tertiles, 4-20); mean: 15.2	Long-term sick leave due to depressive disorder, ≥30 days Employer register ICD-10: F32	Cox proportional hazards model HR (95% CI) PCE: Yes Age, education, marital status, occupation, chronic physical condition 47 / 15 256 **JS: high: 2.22 (1.02-4.83)** med: 1.45 (0.65-3.24) PD: high: 1.01 (0.49-2.04) med: 0.99 (0.49-2.01) **JC: high: 0.27 (0.11-0.67)** med: 0.78 (0.41-1.46) Supervisor SS: high: 1.11 (0.55-2.23) med: 1.21 (0.61-2.40) Co-worker SS: high: 1.01 (0.50-2.05) med: 1.01 (0.51-2.00)
Kivimaki (2010) Register study on overcrowding and antidepressants	Baseline: 2004 FU: 1 y Female nurses of 203 somatic illness ward in 16 hospitals	Exposure at baseline Based on JCQ PD: 3 items, continuous, 1-5 mean: 3.6	1-Absence due to any mental or behavioral disorder, >9 days Government registers ICD-10: F00-F99	1-Conventional probit regression analysis Probit index (95% CI)

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
Finland [26]	2784 / ≈ 3978 PB: 78%, FU: NA, MD: 5% Women only Mean age: 42.2 y, range: 20-64y SES: All nurses		2-Absence due to depressive disorder, >9 days Government registers ICD-10: F32-F34	2- Instrumental probit regression analysis (instrument: overcrowding) Probit index (95% CI) PCE: Yes Age (all women and all nurses, no need to adjust for SES) 1-Any mental or behavioral disorder: 102 / 2784 PD: 1-Conventional **0.176 (0.068-0.285), p=0.0001** 2-Instrumental **0.839 (0.541-1.137), p<0.0001** 2- Depressive disorder: 33 / 2784 PD: 1-Conventional **0.251 (0.078-0.424), p=0.004** 2-Instrumental **0.881 (0.450-1.31), p<0.0001**
Clumeck (2009) Belstress I Belgium[27]	Baseline: 1994-1998 FU: mean: 3y; A subsample of 11 companies from the 24 companies of the Belstress I: large variety of the Belgian work force. 8550 / ≈ 44 623 PB: 48%, FU: ND, MD: 9% 2447 / 6103 Age range: 35-59 y Men: <45 y: 44%; >45 y:56%; Women: <45 y: 58%; >45 y: 42%	Exposure at baseline Based on JCQ, French version [16] JS: quadrant, median split High strain: 24.2% PD: 5 item, tertiles: T3: 30.1% JC: 9 items, tertiles: T1: 35.9% SS: 8 items, tertiles: T1: 40.7% IS: 4 categories, IS: 14.0%	Long-term sick leave due to depression, >28 days Employer register Medico-administrative data Codes: NA	Logistic regression, OR (95% CI) PCE: Unclear Stratified for sex Age, living situation, occupation, All: 200 / 8550 Men: 100 / 6103 Women: 100 / 2447 JS: Men: **Passive: 3.27 (1.41-7.54)** Active: 2.11 (0.88-5.03) **High strain: 4.58 (2.01-10.46)**

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country, (reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
	Occupational level (ISCO): Men: Executives: 21.7%; White-collars: 51%; Blue-collars: 27.3%; Women: Executives: 12.7%; White-collars: 77.2%; Blue-collars: 10.1%			Women: Passive: 1.56 (0.68-3.55) Active: 0.98 (0.40-2.37) High strain: 2.00 (0.90-4.43) PD: Men: high: 1.50 (0.82-2.70) **med: 1.91 (1.12-3.27)** Women: high: 1.17 (0.66-2.07) med: 1.08 (0.63-1.86) **JC: Men: low: 2.83 (1.48-5.42)** med: 1.91 (0.99-3.70) **Women: low: 2.32 (1.10-4.90)** med: 1.03 (0.46-2.29) SS: Men: low: 1.10 (0.63-1.91) med: 0.59 (0.31-1.12) Women: low: 0.98 (0.53-1.80) med: 1.21 (0.65-2.22) **IS: Men: 2.86 (1.58-5.15)** Women: 1.65 (0.93-2.92)
Wannstrom (2009) County council employees in southern Sweden, Kalmar [28]	Baseline: 2002 FU: 44 months after baseline Health workers, majority involved in clinical work 3976 / 6118 PB: 65%, FU: NA, MD: NA 3280 / 696 Age: >48 y: 48.8%; <48 y: 52.2% Education: % with higher education: 53.7%	Exposure at baseline General Nordic Questionnaire for Psychological and Social Factors at Work (QPS_{Nordic}) [28] PD: Quantitative demand: 4 items, mean: 3.06; Decision demand: 3 items, mean: 3.63; Learning demand: 3 items, mean: 2.64 JC: Control of decision: 4 items, mean: 2.83; Control of pacing: 4 items, mean: 2.99	Long-term sick leave due to psychiatric disorder ≥90 days Insurance company database ICD-10 F00-F99	Pearson correlations and point-biserial correlation coefficient PCE: Yes Crude 85 / 3875 **Quantitative demand: 0.045; Decision demand: 0.048; Learning demand 0.086; p<0.01 for each** Control of decision: -0.018; Control of pacing: -0.021; p >0.05 for each

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
		SS: Superior: 3 items, mean:3.42; Co-workers:2 items, mean: 3.97 Score range: NA		**Superior SS: -0.100** **Co-workers SS: -0.113** **p<0.01 for each**
Bourbonnais (2005) Quebec Nurses Study Canada [29]	Baseline: 1997 and 1998, considered as time-varying FU: 17 months from 1997 Nurses of 13 health facilities in Quebec City 1314 / 3152 PB: 70%, FU: 66%, MD: NA 1235 / 79 Mean age: 42 y; range: 22-58 y SES: All nurses	Exposure at two time points (1997 and 1998), considering as time-varying JCQ French version [2] mean or prevalence: NA JS: quadrant PD: 9 items, dichotomized, median split, reference population, JC: 9 items, dichotomized, median split, reference population SS: 8 items, dichotomized, median split Siegrist's questionnaire, mean or prevalence: NA R: 11 items, dichotomized, median split ERI: PD/R ratio, dichotomized: ≤1, >1	Certificated sick leave due to mental disorder, >3 days Employer and insurance registers, and Occupational physicians ICD-9: 300.0, 300.4, 300.5, 300.9, 306.2, 346.9, 427.2, 780.5, 780.7, 994.5 [30]	Cox proportional hazards model, Andersen-Gill method for time-dependant exposure and multiple events HR (95% CI) PCE: Unclear Age, occupational status and previous absence 350 / 1314 (possibility of more than one absence per subject) JS: Passive: 1.78 (0.81-3.91); **Active: 2.28 (1.19-4.36)** **High strain: 2.88 (1.49-5.57)** **PD: 1.93 (1.25-3.00)** JC: 1.39 (0.97-2.02) **SS: 1.74 (1.19-2.55)** **R: 1.96 (1.29-2.99)** **ERI: 1.84 (1.25-2.72)**
Boedeker (2001) Collaborative Program Occupation and Health Germany [31]	Baseline: 1995-1997 FU: 3 y from baseline Employees of five companies from the metal processing and retail trade 42 508 / 42 508 PB: 100%, FU: NA, MD: 0%	Exposure at baseline Job exposure matrix, based on JCQ [32] on 79 job types and 70 workload items Prevalence or mean: NA PD: items: NA, 0-100, five or two categories	Sickness absence for neurotic disorder ≥1 days for 72% and ≥3 days for 28% Insurance registers ICD-9 300–309, 311, 316	Poisson regression RR (95% CI) PCE: Unclear Age, sex, company, education, work hours, and duration of employment

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
	28 905 / 13 603 Median age: Men: 40 y; Women: 42 y SES: NA	JC: items: NA, 0-100, five categories		3159 / 45 508 (possibility of more than one absence per subject) **PD: 0-25: 0.86 (0.81-0.91)** **25-50: 0.85 (0.81-0.89)** **50-75: 0.62 (0.58-0.66)** **>75: 0.44 (0.36-0.53)** **JC: 0-25: 2.17 (1.67-2.83)** **25-50: 2.43 (1.83-3.22)** **50-75: 2.81 (2.16-3.67)** **>75: 3.17 (2.42-4.16)**
Bourbonnais (2001) Quebec Nurses Study Canada [33]	Baseline:1994 FU: May 1994 to Dec 1995, 20 months Nurses employed in six acute care hospitals in the province of Quebec 1793 / 3065 PB: 62%, FU: NA, MD: 5% All women Mean age: 39.6 y; range: 22-65 SES: All nurses	Exposure at baseline JCQ French version [2] Mean or prevalence: NA JS: quadrant PD: 9 items, dichotomized, median split, reference population JC: 9 items, dichotomized, median split, reference population SS: 8 items, dichotomized, median split	Certificated sick leave due to mental disorder, >3-5 days Employer and insurance registers, and occupational physicians ICD-9: 300.0, 300.4, 300.5, 300.9, 306.2, 346.9, 427.2, 780.5, 780.7, 994.5 [30]	Poisson regression RR (95% CI) PCE: Unclear Stratified for unit and number of days worked over 2 weeks adjusted for age and SS at work JS: critical unit care and >10days worked: **High strain: 3.38 (1.55-7.37)** General care unit and <10 days worked: **High strain: 0.27 (0.14-0.54)** Crude: JS: passive: 1.01 (0.55-1.85) Active: 0.90 (0.47-1.70) High strain: 1.24 (0.70-2.21) SS: 1.78 (1.18-2.68) Adjusted for SS only: High strain: 0.99 (0.68-1.46)

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
Stansfeld (1997) Whitehall II United Kingdom [34]	Baseline: 1985-88 FU: to Dec 1990, 5 y White-collar workers, non-industrial civil servants in London, 20 departments 4202 / ≈ 14 121 PB: 73%, FU: 93%, MD: 8% 1116 / 3086 Mean age: NA, range: 35-55y Employment grade: Men: Administrative: 41.3%; Professional/Executive: 53.3% Clerical/support: 5.3% Women: Administrative: 12.9%; Professional/Executive: 44.5%; Clerical/support: 42.6%	Exposure at baseline Whitehall I SS at work [35] SS: 6 items, in tertiles Low social support: Men: 29-40% Women: 33-41% depending on employment grade	Psychiatric sickness absence >7 days Employer registers Mental reasons coded by civil services, validated with a psychiatric diagnosis or a general practitioner for absences of >21 days Codes: NA	Poisson regression RR (95% CI) PCE: Unclear Stratified by sex, adjusted for age, employment grade All: 226 / 4202 Men: 100 / 3086 Women: 126 / 1116 (possibility of more than one absence per subject) **SS:** (ref=low) Men: medium: 0.62 (0.37-1.04) high: 1.01 (0.65-1.59) Women: medium: 1.06 (0.69-1.64), high: 1.09 (0.72-1.65)
Bourbonnais (1995) Quebec white-collar workers Canada [36]	Baseline: 1992-1993 FU: 1992-1994, 1 y White-collar workers in Quebec City 2131 / 4467 PB: 65%, FU: NA, MD: 26% 1023 / 1108 Age: 35-44 y: 45%; 45-54 y: 25%; 15-34 y: 23% Education: Primary or Secondary: 28%; College: 25%; University: 48%	Exposure at baseline JCQ French version [2] Mean or prevalence: NA JS: quadrant PD: 9 items, dichotomized, median split, reference population JC: 9 items, dichotomized, median split, reference population SS: 8 items, dichotomized, median split	Certificated sick leave due to mental disorder Minimal days: NA Government and employer registers ICD Codes: NA	Logistic regression, OR (95% CI) Stratified by social support, stress events and sex PCE : Unclear All: 53 / 2131 Stratified analyses: Men: 10 / 354 Women: 8 / 296 JS: High strain vs low strain men: 0.62 (0.17-2.24) women: 3.03 (0.37-25.0) SS: high: 1.70 (0.53-5.47) low: 0.84 (0.09-7.69) Stressful life events: yes: 1.38 (0.35-5.45)

© 2020 American Medical Association. All rights reserved.

Study **Author (year)** **Cohort name** **Country,** **(reference)**	**Population / Design** **Years, Follow-up (FU)** **Type of workers** **N analyzed / N eligible** **Participation at baseline (PB), missing data (MD)** **N women / N men** **Age, SES**	**Psychosocial stressors at work** **Measurement time** **Tool (reference of validation)** **Number of items** **Prevalence or mean score**	**Sickness absence** **Name** **Minimum absence duration** **Measure** **ICD-10 or 9 codes**	**Analyses / Results** **Model** **Prevalent case excluded (PCE): Yes/No/Unclear** **Covariates** **N cases / n analyzed** **Results**
				no: 0.73 (0.14-3.64 Crude: JS: Passive: 1.10 (0.47-2.60); Active: 0.89 (0.34-2.37); High strain:1.05 (0.40-2.74); SS: low: 0.71 (0.39-1.31)

Results in bold are statistically significant

Abb. SES: socio-economic status; PP: participation proportion; y: year, FU: follow-up, ICD: International Classification of Diseases; NA: not available; JCQ: Job Content Questionnaire; ERI: Effort-reward imbalance; DCSQ: Demand-Control-Support Questionnaire; HR: hazard ratio; OR: odds ratio CI: confidence interval; OJ: organizational justice; RJ: relational justice; PJ: procedural Justice; JS: job strain; PD: psychological demand; JC: job control; SS: social support at work; IS: iso-strain; E: effort; R: reward; ERI: effort-reward imbalance; DP: disability pension; JEM: Job Exposure Matrix; PB: participation at baseline, MD: missing data; Q2: 2nd quartile; Q3: 3rd quartile; Q4: 4th quartile

© 2020 American Medical Association. All rights reserved.

eAppendix 1. Search Strategy for Each Electronic Database, First Run in October 2017 and Updated for Medline, Embase and PsycInfo in Febuary 2019

# line	Medline OVID 2017-10-31	Results
1	(job adj2 (control or security or insecurity or strain or stress or stressor or stressors or demand or demands or demanding)).tw.	6406
2	*workload/ or workload.tw.	26843
3	(work adj2 (stress or stressor or stressors)).tw.	4318
4	**1 or 2 or 3**	**35611**
5	*employment/ or *occupations/	33069
6	employee*.tw.	40359
7	(job or jobs).tw.	54613
8	employment.tw.	49786
9	work.ti.	82280
10	*work/	16386
11	*workplace/	9257
12	(occupation or occupations).tw.	35043
13	workplace*.tw.	34419
14	**OR/5-13**	**278253**
15	exp *sociological factors/px	53840
16	*stress, psychological/	68042
17	*social justice/	5282
18	*reward/ or reward.tw.	33169
19	overcommitment.tw.	212
20	siegrist*.tw. or siegrist j.au.	525
21	karasek*.tw. or karasek r.au. or karasek ra.au.	452
22	skill discretion.tw.	146
23	social support.tw. or *Social Support/	47562
24	(iso-strain or iso strain).tw.	68
25	decision authority.tw.	164
26	decision making.tw. or *Decision Making/	126077
27	(decision adj2 latitude*).tw.	532
28	(demand* adj2 latitude*).tw.	115
29	(demand* adj2 control).tw.	1589
30	(psychological* adj2 demand*).tw.	818
31	(equity or inequities or inequity).tw.	16517
32	(intrinsic adj5 effort*).tw.	127
33	psychosocial.tw.	82945
34	((organizational or organisational or distributive or procedural or interactional or relational) adj5 (justice or injustice)).tw.	1146
35	**OR/15- 34**	**398440**

© 2020 American Medical Association. All rights reserved.

36	(psychiatric* adj2 morbidity).tw.	4217
37	Hospitalisation*.tw. or *Hospitalization/ or Hospitalization*.tw.	154744
38	exp *Somatoform Disorders/	13493
39	(((leave or leaves or absence) and (disability or sickness or sick)) or sick-list*).tw.	11468
40	sick leave/	5263
41	psychotrop*.tw.	17222
42	exp *Psychotropic Drugs/	239600
43	(Psychological disorder or Psychological disorders).tw.	2908
44	*Psychophysiologic Disorders/ep, et, pp, px	4853
45	(Mental disorder or Mental disorders).tw.	33737
46	Mental health.tw.	118036
47	*Mental Health/	18747
48	Anxiolytic*.tw.	14007
49	Anxiety.tw.	161403
50	Antidepress*.tw.	62343
51	Absenteeism.tw.	5047
52	*Anxiety/	36536
53	*Absenteeism/	3973
54	depressive.tw. or *Depressive Disorder/	134394
55	*Depression/ or depression.tw.	311242
56	*Anxiety Disorders/	21425
57	*Mental Disorders/ep, et, pp, px	34852
58	**OR/36-57**	**964423**
59	**4 and 58**	**4331**
60	**14 and 35 and 58**	**5912**
61	**59 or 60**	**8489**
	Results from the same search strategy in Medline OVID 2019-01-29	**9625**
	Additonal citation identified	**1136**

# line	EMBASE 2017-11-01	Results
#1	(job NEAR/2 (control OR security OR insecurity OR strain OR stress OR stressor OR st ressors OR demand OR demands OR demanding)):ab,ti	6 255
#2	'workload'/exp OR workload:ab,ti OR 'work load':ab,ti	48 267
#3	(work NEAR/2 (stress OR stressor OR stressors)):ab,ti	4 508
#4	**#1 OR #2 OR #3**	**56 561**
#5	'employee'/exp/mj OR employee*:ab,ti	45 587
#6	'work'/mj OR work:ti OR job:ab,ti OR jobs:ab,ti	145 851

© 2020 American Medical Association. All rights reserved.

#7	'employment'/mj OR 'occupation'/mj OR employment:ab,ti OR occupation:ab,ti OR occupations:ab,ti	109 131
#8	'workplace'/exp/mj OR workplace:ab,ti	35 960
#9	**#5 OR #6 OR #7 OR #8**	**293 968**
#10	'mental stress'/exp/mj	29 695
#11	'social psychology'/mj	24 422
#12	'social justice'/mj	3 357
#13	'reward'/mj OR reward:ab,ti	34 124
#14	overcommitment:ab,ti	200
#15	'siegrist j.'/au	350
#16	siegrist*:ab,ti	159
#17	'karasek r.'/au OR 'karasek r.a.'/au OR karasek:ab,ti	511
#18	'skill discretion':ab,ti	137
#19	'social support'/mj OR 'social support':ab,ti	42 809
#20	'iso strain':ab,ti OR isostrain:ab,ti	76
#21	'decision authority':ab,ti	172
#22	'decision making'/mj OR 'decision making':ab,ti OR 'decision-making':ab,ti	148 413
#23	(decision NEAR/2 latitude*):ab,ti	520
#24	(demand* NEAR/2 latitude*):ab,ti	119
#25	(demand* NEAR/2 control):ab,ti	1 613
#26	(psychologic* NEAR/2 demand*):ab,ti	890
#27	equity:ab,ti OR inequity:ab,ti OR inequities:ab,ti	15 375
#28	(intrinsic NEAR/5 effort*):ab,ti	112
#29	((organizational OR organisational OR distributive OR procedural OR interactional OR relational) NEAR/5 (justice OR injustice)):ab,ti	1 011
#30	psychosocial:ab,ti	96 085
#31	**#10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30**	**370 441**
#32	'psychiatric morbidity':ab,ti	4 071
#33	'hospitalization'/mj OR hospitalization:ab,ti OR hospitalisation:ab,ti	175 895
#34	somatoform disorder'/exp/mj	9 785
#35	medical leave'/de OR ((leave:ti,ab OR leaves:ti,ab OR absence:ti,ab) AND (disability:ti,ab OR sickness:ti,ab OR sick:ti,ab)) OR 'sick list*':ti,ab	16 001
#36	'psychotropic agent'/exp/mj OR psychotrop*:ab,ti	449 826
#37	'mental disease'/mj OR 'mental health'/mj OR 'mental health':ab,ti OR 'mental disorder':ab,ti OR 'mental disorders':ab,ti OR (psychophysiologic* NEAR/1 disorder*):ab,ti OR (psychologic* NEAR/1disorder*):ab,ti	244 345

© 2020 American Medical Association. All rights reserved.

#38	'anxiety'/mj OR anxiolytic:ab,ti	58 470
#39	'depression'/mj OR depression:ab,ti OR depressive:ab,ti OR antidepress*:ab,ti	457 440
#40	'absenteeism'/mj OR absenteeism:ab,ti	9 680
#41	**#32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40**	**1 303 120**
#42	**#4 AND #43**	**5 686**
#43	**#9 AND #33 AND #43**	**5 455**
#44	**#44 OR #45**	**9 609**
	Results from the same search strategy in Embase 2019-02-05	**11 401**
	Additonal citation identified	**1 792**

# line	PsycInfo-OVID 2017-11-01	Results
1	(workplace* or occupation or occupations or employment or job or jobs or employee*).ab,ti.	181005
2	work.ti.	59467
3	*employment status/	10403
4	*occupations/	6208
5	*personnel/	5230
6	*working conditions/	16561
7	**1 or 2 or 3 or 4 or 5 or 6**	**230302**
8	(job adj2 (control or security or insecurity or strain or stress or stressor or stressors or demand or demands or demanding)).ab,ti.	8288
9	(work adj2 (stress or stressor or stressors)).ab,ti.	4227
10	(work load or workload).ab,ti.	6739
11	occupational stress.ab,ti.	2286
12	((organizational or organisational or distributive or procedural or interactional or relational) adj5 (justice or injustice)).ab,ti.	4265
13	(intrinsic adj5 (effort or efforts)).ab,ti.	138
14	(equity or inequity or inequities).ab,ti.	14102
15	(psychological* adj2 demand*).ab,ti.	638
16	(intrinsic adj5 effort*).ab,ti.	149
17	(demand* adj2 control).ab,ti.	1325
18	(demand* adj2 latitude*).ab,ti.	69
19	(decision adj2 latitude*).ab,ti.	350
20	18 or 19	354
21	(psychological* adj2 demand*).ab,ti.	638
22	(demand* adj2 control).ab,ti.	1325
23	(demand* adj2 latitude*).ab,ti.	69
24	(decision adj2 latitude*).ab,ti.	350

© 2020 American Medical Association. All rights reserved.

25	22 or 23 or 24	1611
26	decision making.ab,ti.	71942
27	decision authority.ab,ti.	124
28	iso-strain.ab,ti.	29
29	iso strain.ab,ti.	29
30	iso-strain.ab,ti.	29
31	social support.ab,ti.	39651
32	skill discretion.ab,ti.	83
33	"siegrist*".ab,ti.	71
34	"karasek*".ab,ti.	334
35	overcommitment.ab,ti.	138
36	reward.ab,ti.	31999
37	psychosocial.ab,ti.	70716
38	(karasek r or karasek r a or karasek robert or karasek robert a).au.	31
39	(siegrist j or siegrist johannes).au.	152
40	*occupational stress/	16870
41	*work load/	1948
42	*"equity (social)"/ or *"equity (payment)"/	2525
43	*decision making/	51924
44	*social support/	23904
45	*rewards/ or *external rewards/ or *internal rewards/ or *monetary rewards/ or *preferred rewards/	9636
46	*social justice/ or *distributive justice/	4271
47	*psychosocial factors/	22527
48	**or/8-47**	**288050**
49	"mental disorder*".ab,ti.	44886
50	mental health.ab,ti.	149843
51	"psychological disorder*".ab,ti.	3850
52	"psychologic disorder*".ab,ti.	25
53	"sick leave*".ab,ti.	1186
54	leave of absence.ab,ti.	91
55	leaves of absence.ab,ti.	33
56	absenteeism.ab,ti.	3459
57	"antidepress*".ab,ti.	35689
58	"anxiolytic*".ab,ti.	6957
59	anxiety.ab,ti.	165786
60	depression.ab,ti.	210650
61	depressive.ab,ti.	90530
62	(hospitalization* or hospitalisation*).ab,ti.	23742
63	(psychiatric* adj2 morbidity).ab,ti.	3143
64	"psychotrop*".ab,ti.	11733

© 2020 American Medical Association. All rights reserved.

65	exp *antidepressant drugs/	31351
66	exp *antihypertensive drugs/	3952
67	*anxiety/	39497
68	*"commitment (psychiatric)"/ or *outpatient commitment/	1510
69	*"depression (emotion)"/	18768
70	*employee absenteeism/	1716
71	*employee leave benefits/	736
72	*psychiatric hospital admission/ or *hospital admission/ or *psychiatric hospitalization/ or *psychiatric hospital readmission/ or *"commitment (psychiatric)"/ or *hospital discharge/ or *psychiatric hospital discharge/	12375
73	"commitment (psychiatric)"/ or outpatient commitment/	1762
74	*hospitalization/	4619
75	*mental disorders/	62256
76	*mental health/	40495
77	exp *sedatives/	15497
78	exp *somatoform disorders/	10198
79	exp *tranquilizing drugs/	38361
82	(((leave or leaves or absence) and (disability or sickness or sick)) or sick-list*).ab,ti.	3828
83	49 or 50 or 51 or 52 or 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68 or 69 or 70 or 71 or 72 or 73 or 74 or 75 or 76 or 77 or 78 or 79	627415
80	82 OR 83	629409
81	**7 and 48 and 84**	**6446**
	Results from the same search strategy in PsycInfo 2019-02-12	**9286**
	Additonal citation identified	**2840**

# line	**Web of Science 2017-10-31**	**Results**
1	TOPIC: (job NEAR/2 (control or security or insecurity or strain or stress or stressor or stressors or demand or demands or demanding))	14 886
2	TOPIC: ("work load" OR workload)	51 383
3	TOPIC: (work NEAR/2 (stress or stressor or stressors))	11 185
4	**#3 OR #2 OR #1**	**74 042**
5	TS=(occupation OR occupations OR employment)	189 160
6	TS=(personnel OR employee* OR worker*)	393 079
7	TOPIC: (job OR jobs)	157 992
8	TITLE: (work)	357 181
9	TOPIC: (workplace*)	59 881
10	**#9 OR #8 OR #7 OR #6 OR #5**	**997 150**
11	TOPIC: (psychosocial)	91 908

© 2020 American Medical Association. All rights reserved.

12	TOPIC: (reward OR rewards)	77 338
13	TOPIC: (overcommitment)	284
14	AUTHOR: (SIEGRIST J OR KARASEK R OR KARASEK RA) OR TOPIC: (siegrist* OR karasek*)	1 592
15	TOPIC: ("skill discretion")	150
16	TOPIC: ("iso strain" OR isostrain OR iso-strain)	321
17	TOPIC: ("decision authority")	240
18	TOPIC: ("decision making" OR "decision-making")	268 413
19	TOPIC: ("iso-strain" OR "iso strain" OR isostrain)	321
20	TOPIC: ("social support")	53 014
21	TOPIC: (decision NEAR/2 latitude*)	1 080
22	TOPIC: (demand* NEAR/2 control)	6 284
23	TOPIC: (psychological* NEAR/2 demand*)	968
24	TOPIC: (equity OR inequity OR inequities)	58 634
25	TOPIC: (intrinsic NEAR/5 effort*)	282
26	TOPIC: ((organizational or organisational or distributive or procedural or interactional or relational) NEAR/5 (justice or injustice))	7 836
27	**#26 OR #25 OR #24 OR #23 OR #22 OR #21 OR #20 OR #19 OR #18 OR #17 OR #16 OR #15 OR #14 OR #13 OR #12 OR #11**	**539 722**
28	TOPIC: (psychiatric* NEAR/2 morbidity)	5 437
29	TS=(anxiety)	207 009
30	TS=("occupational disease*" OR "occupational health")	16 716
31	TOPIC: (hospitalization* OR hospitalisation*)	109 961
32	TS=("sick leave*" OR absenteeism OR ((leave or leaves or absence) AND (disability or sickness or sick)) OR sick-list*)	23 284
33	TS=(antidepress* OR tranquiliz* OR sedative* OR anxiolytic* OR psychotrop*)	104 111
34	TS=("psychologic* disorder*" OR "psychophysiologic* disorder*" OR "mental health" OR "mental disorder*" OR "somatoform disorder*")	186 480
35	#28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34	592 408
36	**#4 AND #35**	**6 196**
37	**#10 AND #27 AND #35**	**7 747**
38	**#36 OR #37**	**11 473**

© 2020 American Medical Association. All rights reserved.

# line	CINAHL 2017-10-31	Results
1	(TI ((job N2 (control or security or insecurity or strain or stress or stressor or stressors or demand or demands or demanding))) OR AB ((job N2 (control or security or insecurity or strain or stress or stressor or stressors or demand or demands or demanding)))) OR (MJ workload OR TI workload OR AB workload) OR (TI (work N2 (stress or stressor or stressors)) OR AB (work N2 (stress or stressor or stressors))	16256
2	(MM employment OR MM "occupations and professions") OR (TI employee* OR AB employee*) OR (TI (job or jobs) OR AB (job or jobs)) OR (TI employment OR AB employment) OR (TI work) OR (MM work) OR (MM "Work Environment") OR (TI (occupation or occupations) OR AB (occupation or occupations)) OR (TI workplace* OR AB workplace*)	133536
3	**(MM "Sociology+/PF") OR (**SU "psychosocial factors" OR TI psychosocial OR AB psychosocial**) OR** (MM "Stress, Psychological") OR (MM "social justice")OR (MJ reward OR TI reward OR AB reward) OR (TI overcommitment OR AB overcommitment) OR ((ZA "siegrist j") or (ZA "siegrist, j") or (ZA "siegrist, j.") or (ZA "siegrist, johannes") or TI siegrist* or AB siegrist*)) OR ((ZA "karasek r") or (ZA "karasek ra") or (ZA "karasek, r.") or (ZA "karasek, robert") or TI karasek* or AB karasek*)) OR (TI skill discretion OR AB skill discretion) OR (TI social support OR AB social support OR MM support, psychosocial) OR (TI (iso strain OR isostrain OR iso-strain) OR (iso strain OR isostrain OR iso-strain)) OR (TI decision authority OR AB decision authority) OR (TI decision making OR AB decision making OR MM decision making) OR (TI (decision N2 latitude*) OR AB (decision N2 latitude*)) OR (TI (demand* N2 latitude*) OR AB (demand* N2 latitude*)) OR (TI demand* N2 control OR AB demand* N2 control) OR (psychological* N2 demand* OR AB psychological* N2 demand*) OR (TI (equity or inequities or inequity) OR AB (equity or inequities or inequity)) OR (TI intrinsic N5 effort* OR AB intrinsic N5 effort*) OR (TI (((organizational or organisational or distributive or procedural or interactional or relational) N5 (justice or injustice))) OR AB (((organizational or organisational or distributive or procedural or interactional or relational) N5 (justice or injustice))))	431972
4	(TI psychiatric* N2 morbidity OR AB psychiatric* N2 morbidity) OR (TI ("occupational disease" OR "occupational diseases") OR AB ("occupational disease" OR "occupational diseases")) OR (MM "Occupational Diseases/EP/ET/PP/PF") OR ((MM "Hospitalization") OR TI (Hospitalization* or Hospitalisation) OR AB (Hospitalization* or Hospitalisation)) OR (MM "Somatoform Disorders+") OR (TI (((leave or leaves or absence) AND (disability or sickness or sick)) OR TI sick-list*) OR AB (((leave or leaves or absence) AND (disability or sickness or sick)) OR TI sick-list*)) OR (MH "Sick Leave") OR (TI psychotrop* OR AB psychotrop*) OR (MM "Psychotropic Drugs+") OR (TI ("Psychological disorder" or "Psychological disorders" or "Psychologic disorder" or	275363

© 2020 American Medical Association. All rights reserved.

	"Psychologic disorders") OR AB ("Psychological disorder" or "Psychological disorders" or "Psychologic disorder" or "Psychologic disorders")) OR (MM "Psychophysiologic Disorders/EP/ET/PP/PF") OR (MM "Psychophysiologic Disorders") OR (TI (Mental disorder or Mental disorders) OR AB (Mental disorder or Mental disorders)) OR (TI "Mental Health" OR AB "Mental Health") OR (TI "Mental Health" OR AB "Mental Health") OR (MM "Mental Health") OR (TI anxiolytic* OR AB anxiolytic*) OR (TI anxiety* OR AB anxiety*) OR (TI antidepress* OR AB antidepress*) OR (TI Absenteeism OR AB Absenteeism) OR (MM "Anxiety") OR (MM "Absenteeism") OR (TI DEPRESSIVE OR AB DEPRESSIVE) OR ((MM "Depression") OR TI depression OR AB depression) OR (MM "Anxiety Disorders") OR (MM "Mental Disorders/EP/ET/PP/PF")	
5	**1 AND 4**	**2136**
6	**2 AND 3 AND 4**	**5286**
7	**5 OR 6**	**6461**

# line	**Sociological Abstracts 2017-11-01**	**Results**
1	(ti(job NEAR/2 (control OR security OR insecurity OR strain OR stress OR stressor OR stressors OR demand OR demands OR demanding)) OR ab(job NEAR/2 (control OR security OR insecurity OR strain OR stress OR stressor OR stressors OR demand OR demands OR demanding)) OR ti(work NEAR/2 (stress OR stressor OR stressors)) OR ab(work NEAR/2 (stress OR stressor OR stressors)) OR ti(workload OR "work load") OR ab(workload OR "work load"))	**5378**
2	(SU.EXACT("Occupations") OR SU.EXACT("Employment") OR SU.EXACT("Work") OR SU.EXACT("Workplaces") OR SU.EXACT("Work Environment") OR SU.EXACT("Work") OR SU.EXACT("SOCIOLOGY OF WORK") OR SU.EXACT("ORGANIZATIONAL SOCIOLOGY")) OR ti((work OR workplace* OR job OR jobs OR occupation OR occupations OR employee* OR employment)) OR ab((workplace* OR job OR jobs OR occupation OR occupations OR employee* OR employment))	**146098**

© 2020 American Medical Association. All rights reserved.

3	((SU.EXACT("Social Factors") OR SU.EXACT("Psychosocial Factors")) OR ti(Psychosocial) OR ab(Psychosocial)) OR (SU.EXACT("Social Support") OR ti("social support") OR ab("social support")) OR (ti("skill discretion") OR ab("skill discretion")) OR SU.EXACT("Distributive Justice" OR "Procedural Justice" OR "Social Justice") OR (ti(reward) OR ab(reward)) OR (SU.EXACT("Reinforcement" OR "Rewards") OR ti(overcommitment) OR ab(overcommitment)) OR (au.Exact("Siegrist, J" OR "Siegrist, Johannes") OR ti(siegrist*) OR ab(siegrist*)) OR (au.Exact("Karasek, Robert" OR "Karasek, Robert Allen" OR "Karasek, Robert A" OR "Karasek, R" OR "Karasek, Robert A, Jr") OR ti(karasek*) OR ab(karasek*)) OR (ti("iso strain" OR "iso strain" OR isostrain) OR ab("iso strain" OR "iso strain" OR isostrain)) OR (ti("decision authority") OR ab("decision authority") OR ti(decision* NEAR/2 latitude*) OR ab(decision* NEAR/2 latitude*) OR ti(demand* NEAR/2 control) OR ab(demand* NEAR/2 control) OR ti(demand* NEAR/2 latitude*) OR ab(demand* NEAR/2 latitude*) OR ti(psychological* NEAR/2 demand*) OR ab(psychological* NEAR/2 demand*)) OR (SU.EXACT("Equity" OR "Educational Inequality" OR "Income Inequality" OR "Inequality" OR "Sexual Inequality" OR "Social Inequality") OR ti(equity OR inequity OR inequities) OR ab(equity OR inequity OR inequities)) OR (ti((organizational OR organisational OR distributive OR procedural OR interactional OR relational) NEAR/5 (justice OR injustice)) OR ab((organizational OR organisational OR distributive OR procedural OR interactional OR relational) NEAR/5 (justice OR injustice)))	79790
4	(ti(psychiatric NEAR/2 morbidity) OR ab(psychiatric NEAR/2 morbidity)) OR (SU.EXACT("Psychological Stress") OR SU.EXACT("Psychological Factors")) OR **(ti(leave or leaves or absence) AND (all(disability) or all(sickness) or all(sick))) OR (ti(sick-list*)) OR (ab(leave or leaves or absence) AND (all(disability) or all(sickness) or all(sick))) OR (ab(sick-list*))** OR (ti(psychotro*) OR ab(psychotro*) OR (SU.EXACT("Medications") OR SU.EXACT("Tranquilizing Drugs"))) OR (SU.EXACT("Hospitalization") OR ti(hospitalization OR hospitalisation) OR ab(hospitalization OR hospitalisation)) OR (ti(psychotrop*) OR ab(psychotrop*) OR SU.EXACT("Medications") OR SU.EXACT("Tranquilizing Drugs")) OR ((SU.EXACT("Mental Illness") OR ti("mental disorder*") OR ab("mental disorder*")) OR (SU.EXACT("Mental Health") OR ti("mental health") OR ab("mental health"))) OR OR (SU.EXACT("Psychopathology") OR ti(psychopathol*) OR ab(psychopathol*)) OR (ti("psychological disorder*") OR ab("psychological disorder*")) OR (ti("psychologic* disorder*") OR ab("psychologic* disorder*")) OR (SU.EXACT("Anxiety") OR ti(anxiety) OR ab(anxiety)) OR (ti(anxiolytic*) OR ab(anxiolytic*) OR ti(Antidepress*) OR ab(Antidepress*)) OR (SU.EXACT("Depression (Psychology)") OR ti(depression OR depressive) OR ab(depression OR depressive)) OR (SU.EXACT("Absenteeism") OR ti(Absenteeism) OR ab(Absenteeism))	54598

© 2020 American Medical Association. All rights reserved.

5	1 AND 4	**677**
6	2 AND 3 AND 4	**931**
7	**5 OR 6**	**1400**

# line	IBSS 2017-11-02	Results
1	ti((job NEAR/2 (control OR security OR insecurity OR strain OR stress OR stressor OR stressors OR demand OR demands OR demanding)) OR (work NEAR/2 (stress OR stressor OR stressors)) OR (workload OR "work load") OR work) OR ab((job NEAR/2 (control OR security OR insecurity OR strain OR stress OR stressor OR stressors OR demand OR demands OR demanding)) OR (work NEAR/2 (stress OR stressor OR stressors)) OR (workload OR "work load"))	69002
2	(SU.EXACT("Occupations") OR SU.EXACT("Work place") OR SU.EXACT("Work environment") OR SU.EXACT("Workers")) OR ti(work OR workplace* OR employee* OR employment OR job OR jobs OR occupation OR occupations OR worker*) OR ab(work OR workplace* OR employee* OR employment OR job OR jobs OR occupation OR occupations OR worker*)	421039
3	((SU.EXACT("Sociology of work") OR SU.EXACT("Industrial sociology")) OR (SU.EXACT("Social factors") OR SU.EXACT("Psychosociology")) OR SU.EXACT("Social support") OR ti(psychosocial OR "social support") OR ab(psychosocial OR "social support")) OR (ti("skill discretion") OR ab("skill discretion")) OR ((SU.EXACT("Distributive justice") OR SU.EXACT("Social justice")) OR ti((organizational OR organisational OR distributive OR procedural OR interactional OR relational) NEAR/5 (justice OR injustice)) OR ab((organizational OR organisational OR distributive OR procedural OR interactional OR relational) NEAR/5 (justice OR injustice))) OR (ti(reward OR overcommitment) OR ab(reward OR overcommitment)) OR (au.Exact("Siegrist, J" OR "Siegrist, Johannes") OR ti(siegrist*) OR ab(siegrist*)) OR (au.Exact("Karasek, Robert" OR "Karasek, R") OR ti(karasek*) OR ab(karasek*)) OR (ti("iso strain" OR "iso strain" OR isostrain) OR ab("iso strain" OR "iso strain" OR isostrain)) OR (ti("decision authority") OR ab("decision authority")) OR (SU.EXACT("Decision making") OR ti("decision making" OR "decision authority") OR ab("decision making" OR "decision authority")) OR (ti((decision* NEAR/2 latitude*) OR (demand* NEAR/2 control) OR (demand* NEAR/2 latitude*) OR (psychological* NEAR/2 demand*)) OR ab((decision* NEAR/2 latitude*) OR (demand* NEAR/2 control) OR (demand* NEAR/2 latitude*) OR (psychological* NEAR/2 demand*))) OR ((SU.EXACT("Inequality") OR SU.EXACT("Income inequality") OR SU.EXACT("Economic inequality") OR SU.EXACT("Social inequality") OR SU.EXACT("Equity")) OR ti(equity OR inequity OR inequities) OR ab(equity OR inequity OR inequities))	185725

© 2020 American Medical Association. All rights reserved.

4	SU.EXACT("Mental stress") OR	68517
	ti(psychiatric NEAR/2 morbidity) OR	
	ab(psychiatric NEAR/2 morbidity) OR	
	SU.EXACT("Psychological factors") OR	
	SU.EXACT("Hospitalization") OR	
	ti(hospitalization OR hospitalisation) OR	
	ab(hospitalization OR hospitalisation) OR	
	ti(((leave or leaves or absence) and (disability or sickness or sick)) or sick-list*) OR	
	ab(((leave or leaves or absence) and (disability or sickness or sick)) or sick-list*) OR	
	ti(psychotrop* OR medication* OR tranquiliz*) OR	
	ab(psychotrop* OR medication* OR tranquiliz*) OR	
	SU.EXACT("Psychological factors") OR	
	SU.EXACT("Mental health") OR	
	SU.EXACT("Mental stress") OR	
	SU.EXACT("Psychopathology") OR	
	SU.EXACT("Psychological effects") OR	
	SU.EXACT("Mental illness") OR	
	ti("mental disorder*" OR "mental health" OR "mental illness" OR psychopathol* OR "psychological disorder*") OR	
	ab("mental disorder*" OR "mental health" OR "mental illness" OR psychopathol* OR "psychological disorder*") OR	
	SU.EXACT("Anxiety") OR	
	SU.EXACT("Depression") OR	
	ti(anxiety OR depression OR depressive OR anxiolytic* OR antidepress*) OR	
	ab(anxiety OR depression OR depressive OR anxiolytic* OR antidepress*) OR	
	SU.EXACT("Absenteeism") OR	
	ti(Absenteeism) OR	
	ab(Absenteeism)	
5	**1 AND 4**	2805
6	**2 AND 3 AND 4**	1763
7	**5 OR 6**	3996

© 2020 American Medical Association. All rights reserved.

eAppendix 2. Completed ROBINS-I Tool Assessment Grid

The Risk Of Bias In Non-randomized Studies – of Interventions (ROBINS-I) assessment tool

(version for cohort-type studies)

Version 19 September 2016

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

ROBINS-I tool (Stage I): At protocol stage

Specify the review question

Participants	Working adults of all types, from any country. Must not be a population of workers who are off work for illness, who are in a process of return to work or a population of pregnant women.
Experimental intervention	**Type:** Exposure must have been measured with a validated tool or a proxy of a validated tool based on one of three models studied, namely the Job Demand-Control-Support (JDCS), the Effort-Reward imbalance (ERI), and the Organisational justice models. The validity of this tool must have been demonstrated in a study on the psychometric qualities of the instrument (internal consistency, factorial validity, predictive validation and discriminant validity). If a translated version was used, it must also be validated. **Frequency**: Participants must have been exposed for a sufficient amount of time to allow the occurrence of the outcome. In the case of absences for mental disorders, we consider one year of exposure to be enough. Considering that almost all studies measured exposure at a single time point, participants had to have been employed for at least one year in the same organization. **Consideration of past exposure:** To consider previous exposure, a cohort must be 1) comprised of new workers never before exposed or 2) evaluate the change in the exposure from a known past exposure. The consideration of past exposure is difficult in this field of research. **Measurement time:** The exposure must have been measured at the beginning of the study, without participants knowing the nature of the study. **Healthy Cohort in Exposure Measurement**: Participants should not have a mental disorder when measuring exposure (*i.e.* exposure evaluators are not influenced by knowledge of the outcome). Prevalent cases must be excluded.
Comparator	Exposed and unexposed workers originate from the same study population.
Outcomes	**Absences:** Information collected in objective administrative files. The people who collect the information should not be aware of the exposure level of the participants. Information must have been collected in the same way for both exposure groups.

© 2020 American Medical Association. All rights reserved.

	Diagnostics that cause the absence: The diagnosis must have been made by a physician and should not be self-reported by the participants. The physician should not be aware of the exposure level of participants.

List the confounding domains relevant to all or most studies

Major confounding domains (for which we want the analyses to be compulsorily adjusted): Socio-economic status (ideally education or income, but we also accept occupation), Age and Sex.
Additional confounding domains, but optional: Work Environment Factors, Family Charge, Stressful Event, Social support outside work
Intermediate domains (should not be adjusted for or should be included in a separate statistical model): Lifestyle habits (smoking, alcohol, and physical activity), Comorbidities (cardiovascular disease, diabetes, musculoskeletal disorders, etc.) and Body mass index (BMI).

~~List co-interventions that could be different between intervention groups and that could impact on outcomes~~

Non-Applicable

Specify the outcome

Specify which outcome is being assessed for risk of bias (typically from among those earmarked for the Summary of Findings table). Specify whether this is a proposed benefit or harm of intervention.

Outcome: Sickness absence due to a mental disorder as diagnosed by a doctor and certified by a doctor as the reason for the work absence. All diagnoses must be coded according to ICD-10 (F00-F99) or ICD-9 (300-316).

Specify the numerical result being assessed

In case of multiple alternative analyses being presented, specify the numeric result (e.g. RR = 1.52 (95% CI 0.83 to 2.77) and/or a reference (e.g. to a table, figure or paragraph) that uniquely defines the result being assessed.

Hazard ratio (HR), Odds ratio (OR), Risk ratio (RR), Rate ratio (Poisson regression), minimally adjusted for age, sex and SES, with prevalent cases excluded when possible and with individual level exposure instead of aggregated level.

© 2020 American Medical Association. All rights reserved.

Preliminary consideration of confounders

Complete a row for each important confounding domain (i) listed in the review protocol; and (ii) relevant to the setting of this particular study, or which the study authors identified as potentially important.

"Important" confounding domains are those for which, in the context of this study, adjustment is expected to lead to a clinically important change in the estimated effect of the intervention. "Validity" refers to whether the confounding variable or variables fully measure the domain, while "reliability" refers to the precision of the measurement (more measurement error means less reliability).

(i) Confounding domains listed in the review protocol				
Confounding domain	Measured variable(s)	Is there evidence that controlling for this variable was unnecessary?*	Is the confounding domain measured validly and reliably by this variable (or these variables)?	OPTIONAL: Is failure to adjust for this variable (alone) expected to favour the experimental intervention or the comparator?
MAJOR:			Yes / No / No information	Favour experimental / Favour comparator / No information
Age	Self-reported or obtained by register, ideally as a continuous variable or in at least 3 categories.	Never		
Sex	Self-reported or obtained by register, dichotomous.	Population composed solely of women or solely of men		
Socioeconomic status	Education, income or occupation, self-reported or obtained by register, treated as a categorical variable (at least 3 categories).	Population composed of one type of workers only, e.g. nurses		
OPTIONAL:				
Work Environment Factors				

© 2020 American Medical Association. All rights reserved.

Family Charge				
Stressful Event				
Out of Work Social Support				
MEDIATOR (should not be adjusted for)				
Lifestyle habits (alcohol, smoking, physical activity)				
Comorbidity (cardiovascular disease, diabetes, physical illness, self-rated health, musculoskeletal problem, etc.)				
BMI				

* In the context of a particular study, variables can be demonstrated not to be confounders and so are not included in the analysis: (a) if they are not predictive of the outcome; (b) if they are not predictive of intervention; or (c) because adjustment makes no or minimal difference to the estimated effect of the primary parameter. Note that “no statistically significant association” is not the same as “not predictive”.

© 2020 American Medical Association. All rights reserved.

Risk of bias assessment

Responses underlined in green are potential markers for low risk of bias, and responses in red are potential markers for a risk of bias. Where questions relate only to sign posts to other questions, no formatting is used.

	Signalling questions	Description	Response options
Bias due to confounding			
	1.1 Is there potential for confounding of the effect of intervention in this study? ~~**If N/PN to 1.1:** the study can be considered to be at low risk of bias due to confounding and no further signalling questions need to be considered~~	**YES**: The answer will always be YES in observational studies.	**Y** ~~/ PY~~ / ~~PN / N~~
	If Y/PY to 1.1: determine whether there is a need to assess time-varying confounding:		
	1.2. Was the analysis based on splitting participants' follow-up time according to intervention received? **If N/PN**, answer questions relating to baseline confounding (1.4 to 1.6) **If Y/PY**, go to question 1.3.	**NO:** If exposure is measured at recruitment only, or if exposure is measured at two or more times, but the authors computed an average exposure over measurement times. **YES:** If the exposure is measured at several times and considered cumulatively or combined.	NA / Y / PY / PN / N / NI
	1.3. Were intervention discontinuations or switches likely to be related to factors that are prognostic for the outcome? ~~**If N/PN**, answer questions relating to baseline confounding (1.4 to 1.6)~~ **If Y/PY**, answer questions relating to both baseline and time-varying confounding (1.7 and 1.8)	**IF YES at question 1.2:** **YES:** Always yes. It can always be assumed that the change in exposure status may be influenced by the presence of depressive symptoms. **IF NO at question 1.2: NA**	NA / Y ~~/ PY / PN / N /~~ NI

	Questions relating to baseline confounding only

© 2020 American Medical Association. All rights reserved.

	1.4. Did the authors use an appropriate analysis method that controlled for all the important confounding domains?	Appropriate analyses: adjustment, restriction, weighting, backward change in estimate procedure Not appropriate: Forward change in estimate procedure, one factor at a time in the crude model Significant domains of confusion: age, sex, socio-economic status (SES) **YES**: if appropriate analysis for age, sex and SES **NO**: if any of these factors were not considered or not considered appropriately in the analysis	NA / Y / PY / PN / N / NI
	1.5. **If Y/PY to 1.4**: Were confounding domains that were controlled for measured validly and reliably by the variables available in this study?	Age: by register or self-reported, continuous or with at least 3 categories Sex: by register or self-reported SSE: Self-reported for education and occupation are considered acceptable. Register for occupation is acceptable, income: self-reported suboptimal, but there is not really any other source of information possible. Therefore we consider self-reported income acceptable. Should be reported with at least 3 categories **YES:** if all three measures are relatively valid. **NO:** if one of the measures seems very invalid or poorly measured or misclassified (e.g. age reported dichotomously)	NA / Y / PY / PN / N / NI
	1.6. Did the authors control for any post-intervention variables that could have been affected by the intervention?	Potentially Intermediate Variables: Lifestyle, BMI, and Comorbidities **NO**: if there is a model with appropriate analyses for age, sex and SES, *without* potentially intermediate variables **YES**: if there are no model with appropriate analyses for age, sex and SES, without potentially intermediate variables	NA / Y / PY / PN / N / NI
	Questions relating to baseline and time-varying confounding		
	1.7. Did the authors use an appropriate analysis method that controlled for all the important confounding domains and for time-varying confounding?	**NO:** will almost always be no, because the appropriate analysis would be marginal structural models and to our knowledge it has not yet been done in our field of research. **YES:** Marginal structural models or Cox model with exposure and confounders considered time-varying, with point and non-combined or cumulative exposure	NA / Y / PY / PN / N / NI

© 2020 American Medical Association. All rights reserved.

	Risk of bias judgment	**Low:** No confounding expected: **NEVER** **Moderate**: (i) Confounding expected, all known important confounding domains appropriately measured and controlled for; *and* (ii) Reliability and validity of measurement of important domains were sufficient, such that we do not expect serious residual confounding. **Serious**: (i) At least one known important domain was not appropriately measured, or not controlled for; *or* (ii) Reliability or validity of measurement of an important domain was low enough that we expect serious residual confounding. **Critique:** (i) Confounding inherently not controllable *or* (ii) The use of negative controls strongly suggests unmeasured confounding.	~~Low~~ / Moderate / Serious / Critical / NI

Bias in selection of participants into the study		
2.1. Was selection of participants into the study (or into the analysis) based on participant characteristics observed after the start of intervention? **~~If N/PN to 2.1:~~** ~~go to 2.4~~	**YES:** Always yes in our case, except with a cohort of new workers or have selected a cohort of participants who would all be exposed, or all not exposed at recruitment, and analyze the change in exposure over time.	Y / PY / ~~PN / N~~ / NI
2.2. **If Y/PY to 2.1**: Were the post-intervention variables that influenced selection likely to be associated with intervention?	**YES:** Always yes, because we can expect more exposed people to leave work before the start of the study, or to participate less in the study	NA / Y / PY / ~~PN / N~~ / NI
2.3 **If Y/PY to 2.2**: Were the post-intervention variables that influenced selection likely to be influenced by the outcome or a cause of the outcome?	**YES:** Always yes, because we can expect more exposed people to leave work before the start of the study, or to participate less in the study.	NA / Y / PY / ~~PN / N~~ / NI
2.4. Do start of follow-up and start of intervention coincide for most participants?	**NO:** Due to our field of study, this answer should always be no, except with a cohort of new workers or having selected a cohort of participants who would all be exposed or all unexposed at recruitment, and analyzed the change in exposure over time.	~~Y / PY~~ / PN / N / NI
2.5. **If Y/PY to 2.2 and 2.3~~, or N/PN to 2.4~~**: Were adjustment techniques used that are likely to correct for the presence of selection biases?	**NO:** Always no, because we never know the characteristics of the participants before the start of the study.	NA / ~~Y / PY~~ / PN / N / NI

© 2020 American Medical Association. All rights reserved.

Risk of bias judgment	**Note:** In occupational studies, start of follow-up and start of exposure rarely coincide. For this reason, we choose to start the risk of bias in selection of participants into the study to a moderate level. However, in order to increase discrimination property in this risk of bias, the criterion of the coincidence between start of follow-up and start of intervention will not be further considered. **Low:** **Never, as explained in the note above.** (i) All participants who would have been eligible for the target trial were included in the study; *and* (ii) For each participant start of follow up and start of intervention coincided. **Moderate:** **Participation rates of ≥80% or ≥70% with a comparison showing that refusals are similar to those included for age, sex and socio-economic status, or for exposure and outcome** (i) Selection into the study may have been related to intervention and outcome; *and* The authors used appropriate methods to adjust for the selection bias; *or* (ii) ~~Start of follow up and start of intervention does not coincide for all participants; *and* (a)~~ the proportion of participants for which this was the case was too low to induce important bias; (80%) ~~*or* (b) the authors used appropriate methods to adjust for the selection bias; *or* (c) the review authors are confident that the rate (hazard) ratio for the effect of intervention remains constant over time.~~ **Serious:** **Participation rates between 80-60% or 60%-50% with a comparison showing that refusals are similar to those included for age, sex and socio-economic status, or for exposure and outcome** (i) Selection into the study was related (but not very strongly) to intervention and outcome; *and* This could not be adjusted for in analyses; *or* (ii) ~~Start of follow up and start of intervention does not coincide; *and*~~ A potentially important amount of follow-up time is missing from analyses; *and* The rate ratio is not constant over time. **Critical:** **Participation rates of less <60% or <50% with a comparison showing that refusals are similar to those included for age, sex and socio-economic status, or for exposure and outcome** (i) Selection into the study was very strongly related to intervention and outcome; *and* This could not be adjusted for in analyses; or (ii) A substantial amount of follow-up time is likely to be missing from analyses; *and* The rate ratio is not constant over time.	~~Low /~~ Moderate / Serious / Critical / NI

© 2020 American Medical Association. All rights reserved.

Bias in classification of interventions		
3.1 Were intervention groups clearly defined?	**YES:** Exposure must have been measured by a validated tool based on one of three models studied. The validity must have been demonstrated in a study on the psychometric qualities of the instrument (internal consistency, factorial validity, predictive validation and discriminant validity). Note: If the tool used is an original validated tool, but the translation has not been validated, it is considered to be a well-defined intervention, but with a moderate level of risk. **NO:** Exposure measured with a proxy or translation whose validation has not been demonstrated, or by using different questionnaires from one participant to another. Exposure measured by a matrix based on job titles or based on the response of colleagues in the same work unit, as there is a risk of significant misclassification.	Y / PY / PN / N / NI
3.2 Was the information used to define intervention groups recorded at the start of the intervention?	**YES**: If exposure is measured at beginning of follow-up **NO**: If exposure is measured retrospectively	Y / PY / PN / N / NI
3.3 Could classification of intervention status have been affected by knowledge of the outcome or risk of the outcome?	**NO:** If prevalent cases were excluded **PY:** If the analyses are adjusted for mental health to recruitment only without the exclusion of prevalent cases **YES:** If prevalent cases are not excluded and no sensitivity analysis was conducted according to the mental health status at recruitment. In this case, the disease may have affected the response to the exposure questions.	Y / PY / PN / N / NI

© 2020 American Medical Association. All rights reserved.

Risk of bias judgment	**Low:** (i) Intervention status is well defined; *and* (ii) Intervention definition is based solely on information collected at the time of intervention. **Moderate:** (i) Intervention status is well defined; (Note: here we included the use of validated questionnaire, but without validation of the translation) *and* (ii) Some aspects of the assignments of intervention status were determined retrospectively. **Serious:** (i) Intervention status is not well defined; *or* (ii) Major aspects of the assignments of intervention status were determined in a way that could have been affected by knowledge of the outcome. **Critical:** (Unusual) An extremely high amount of misclassification of intervention status, e.g. because of unusually strong recall biases.	Low / Moderate / Serious / Critical / NI

Bias due to deviations from intended interventions: NA: Hard to apply in our field of research. Exposure deviations are almost always natural and expected, unless there is an intervention by a researcher that is differential depending on the level of exposure. This criterion will always be at a moderate level of risk. Therefore, it is not systematically evaluated in the included studies.

Bias due to missing data: **<u>NOTE:</u> This bias domain concerned participation in the follow-up and number and processing of missing data. The rate of participation in recruitment has been already taken into account in the selection into the study bias domain.**			
	5.1 Were outcome data available for all, or nearly all, participants?	**YES**: If participation at follow-up is 95% and over and /or data is complete for 95% of participants. Data complete for 90% of participants with a comparison between those included and excluded showing similarity for the three confounders important or for exposure and for the outcome will be considered adequate. **NO:** If less than 90% of participants are included in the analysis or less than 95% without comparison or with a comparison showing differences	<u>Y / PY</u> / PN / N / NI

© 2020 American Medical Association. All rights reserved.

	5.2 Were participants excluded due to missing data on intervention status?	**YES**: If there were missing data on exposure **NO:** If there are no missing data on the exposure	Y / PY / PN / N / NI
	5.3 Were participants excluded due to missing data on other variables needed for the analysis?	**YES:** If there were missing data on covariates **NO:** If there are no missing data on covariates	Y / PY / PN / N / NI
	5.4 **If PN/N to 5.1, or Y/PY to 5.2 or 5.3**: Are the proportion of participants and reasons for missing data similar across interventions?	**YES:** If there is a comparison between included and excluded participants due to missing data that shows participants are similar for all three important confounders or for exposure and outcome **NO:** If the comparison shows that the included and excluded are different **NI:** If no information is provided on the differences between the included and the excluded	NA / Y / PY / PN / N / NI
	5.5 **If PN/N to 5.1, or Y/PY to 5.2 or 5.3**: Is there evidence that results were robust to the presence of missing data?	**YES:** If a sensitivity analysis was performed to account for missing data (multiple imputation, inverse probability weighting) and the results are similar to the main analysis, or the results are different but the interpretation is done on the sensitivity analysis and not on the main analysis. **NO:** If no sensitivity analysis is done for missing data	NA / Y / PY / PN / N / NI

© 2020 American Medical Association. All rights reserved.

	Risk of bias judgment	**Low:** (i) Data were reasonably complete; (**95% or 90% with demonstrations that they are similar or an analysis was done for missing data**) *or* (ii) Proportions of and reasons for missing participants were similar across intervention groups; *or* (iii) The analysis addressed missing data and is likely to have removed any risk of bias. **Moderate (between 94 (or 89) and 80% at follow-up, can go down to 75% if a comparison shows that they are similar):** (i) Proportions of and reasons for missing participants differ slightly across intervention groups; *and* (ii) The analysis is unlikely to have removed the risk of bias arising from the missing data. **Serious (between 79% (or 74%) and 50% at follow-up with comparison):** (i) Proportions of missing participants differ substantially across interventions; *or* Reasons for missing differ substantially across interventions; *and* (ii) The analysis is unlikely to have removed the risk of bias arising from the missing data; *or* Missing data were addressed inappropriately in the analysis; *or* The nature of the missing data means that the risk of bias cannot be removed through appropriate analysis. **Critical (<50%):** (i) (Unusual) There were critical differences between interventions in participants with missing data; *and* (ii) Missing data were not, or could not, be addressed through appropriate analysis	Low / Moderate / Serious / Critical / NI

© 2020 American Medical Association. All rights reserved.

Bias in measurement of outcomes			
	6.1 Could the outcome measure have been influenced by knowledge of the intervention received?	**NO:** Obtained objectively by register **PN:** some of the motives for absences are obtained by the interview of participants **YES:** impossible in our case due to inclusion criteria's and the definition of the chosen issue.	Y / PY / PN / N / NI
	6.2 Were outcome assessors aware of the intervention received by study participants?	**NO:** If the people responsible for collecting data in the register do not know the status of the exhibition **YES**: very unlikely, only if the persons responsible for collecting data in the registers know the status of the exhibition	Y / PY / PN / N / NI
	6.3 Were the methods of outcome assessment comparable across intervention groups?	**Always YES,** unless the method is different between exposed and unexposed, which would be highly unlikely	Y / PY / PN / N / NI
	6.4 Were any systematic errors in measurement of the outcome related to intervention received?	**Always NO:** if there are errors in the outcome measure they will not be related to the exposure	Y / PY / PN / N / NI

© 2020 American Medical Association. All rights reserved.

	Risk of bias judgment	**Low:** (i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Moderate:** (i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure is only minimally influenced by knowledge of the intervention received by study participants**;** *and* (iii) Any error in measuring the outcome is only minimally related to intervention status. **Serious: NEVER** (i) The methods of outcome assessment were not comparable across intervention groups; *or* (ii) The outcome measure was subjective (i.e. vulnerable to influence by knowledge of the intervention received by study participants); *and* The outcome was assessed by assessors aware of the intervention received by study participants; *or* (iii) Error in measuring the outcome was related to intervention status. **Critical: Never**	Low / Moderate / ~~Serious /~~ ~~Critical~~ / NI

Bias in selection of the reported result:
Note: This criterion is difficult to apply in our case, because no (or very few) studies have a published protocol and we never have access to the analysis plan. Thus, would always be at moderate risk, therefore will not be evaluated systematically in included studies

© 2020 American Medical Association. All rights reserved.

Overall - SUMMARY			
	Bias due to confounding		~~Low~~ / Moderate / Serious / Critical / NI
	Bias in selection of participants into the study		~~Low /~~ Moderate / Serious / Critical / NI
	Bias in classification of interventions		Low / Moderate / Serious / Critical / NI
	Bias due to missing data		Low / Moderate / Serious / Critical / NI
	Bias in measurement of outcomes		Low / Moderate / ~~Serious~~ / ~~Critical~~ / NI
	Overall		

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

© 2020 American Medical Association. All rights reserved.

eAppendix 3. Quality Evaluation of Prospective Studies on Psychosocial Stressors at Work and Sickness Absence Due to a Diagnosed Mental Disorder, According to Risk of Bias in Non-randomised Studies-Intervention Tool (ROBINS-I) Criteria

Roelen 2018		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for; *and* (ii) Reliability and validity of measurement of important domains were sufficient, such that we do not expect serious residual confounding. **Adjusted for age, sex, marital status, care for children and work related variables (type of nursing work, years as a nurse, work hours/week) obtained from a register**
Bias in selection of participants into the study	Critical	(ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation at baseline was 38% without comparison between participants and non-participants**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **Validated questionnaire, but no reference provided for the validation of the translation** *and* (ii) Some aspects of the assignments of intervention status were determined retrospectively. **The authors do not mention if the prevalent cases were excluded at baseline.**
Bias due to missing data	Serious	(i) Proportions of missing participants differ substantially across interventions; *or* Reasons for missing differ substantially across interventions; *and* (ii) The analysis is unlikely to have removed the risk of bias arising from the missing data; *or* Missing data were addressed inappropriately in the analysis; **74% of baseline participants were included in the analyses and authors provide comparison between included and excluded participants showing that lost to follow-up are similar in terms of exposure, SES, age and sex. No imputation was done.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	Critical	

© 2020 American Medical Association. All rights reserved.

Vendrig 2018		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Critical	(i) Confounding inherently not controllable: **Confounding not evaluated, only crude results presented**
Bias in selection of participants into the study	No information	**Participation at baseline not mentioned**
Bias in classification of interventions	Low	(i) Intervention status is well defined; *and* (ii) Intervention definition is based solely on information collected at the time of intervention. **Exposure was measured with a validated tool and prevalent cases were excluded at baseline**
Bias due to missing data	No information	**Percent of missing data not mentioned**
Bias in measurement of outcomes	Moderate	(ii) The outcome measure is only minimally influenced by knowledge of the intervention received by study participants**: We do not have the confirmation that absence data came from a register and thus were objectively assessed, but the diagnoses were given by a physician.**
Overall	Critical	

© 2020 American Medical Association. All rights reserved.

van Hoffen 2018		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for; *and* (ii) Reliability and validity of measurement of important domains were sufficient, such that we do not expect serious residual confounding. **Adjusted for age, sex, and education and other work characteristics in a second model**
Bias in selection of participants into the study	Moderate	ii) A potentially important amount of follow-up time is missing from analyses: **Participation of 82% at baseline, without comparison with non-participants**
Bias in classification of interventions	Low	(i) Intervention status is well defined; *and* (ii) Intervention definition is based solely on information collected at the time of intervention. **Exposure was measured with a validated tool and participants on sick leave were excluded at baseline and an adjustment was made for mental health at baseline**
Bias due to missing data	Serious	i) Proportions of missing participants differ substantially across interventions: **69% of the baseline participants were included in the analyses,** Reasons for missing differ substantially across interventions: **excluded workers were younger, more often male, had shorter employment duration, shorter job tenure, and reported more favorable psychosocial work characteristics than included workers**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	Serious	

© 2020 American Medical Association. All rights reserved.

Ndjaboué 2017		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(i) At least one known important domain was not appropriately measured, or not controlled for: **Models are adjusted for age, gender and education (SES) but also include smoking, BMI, sedentary behavior and alcohol intake, which could be considered mediators.**
Bias in selection of participants into the study	Moderate	ii) A potentially important amount of follow-up time is missing from analyses: **Participation of 81% at baseline, without comparison with non-participants**
Bias in classification of interventions	Low	(i) Intervention status is well defined; *and* (ii) Intervention definition is based solely on information collected at the time of intervention. **Exposure was measured with a validated tool and prevalent cases were excluded at baseline**
Bias due to missing data	Low	(i) Data were reasonably complete: **Data were at least 95% complete.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	Serious	

© 2020 American Medical Association. All rights reserved.

Otha 2017		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(i) At least one known important domain was not appropriately measured, or not controlled for: **Models are adjusted for age, sex and job position (SES) but include WAI score (subjective estimation of work ability, number of diagnosed diseases, subjective estimation for work impairment due to diseases, sickness absences during the past year, mental resources), which could partly be considered as a mediator.**
Bias in selection of participants into the study	Moderate	(i) Selection into the study may have been related to intervention and outcome; *and* the proportion of participants for which this was the case was too low to induce important bias: **Participation at baseline: 94%**
Bias in classification of interventions	Moderate	(ii) Some aspects of the assignments of intervention status were determined retrospectively: **Prevalent cases not clearly excluded, but adjusted for in the analysis**
Bias due to missing data	Moderate	(i) Proportions of and reasons for missing participants differ slightly across intervention groups; **84% of baseline participants were included in the analyses and authors do not provide comparison between included and excluded participants**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	Serious	

© 2020 American Medical Association. All rights reserved.

Junavi 2016		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for; *and* (ii) Reliability and validity of measurement of important domains were sufficient, such that we do not expect serious residual confounding. **Adjusted for age, sex, location of workplace, occupational status, education and size of residence**
Bias in selection of participants into the study	Serious	(ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation at baseline was 68% without comparison between participants and non-participants**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **Validated questionnaire, but no reference provided for the validation of the translation**
Bias due to missing data	Serious	(i) Proportions of missing participants differ substantially across interventions: **70% of the baseline participants included in the analysis, without comparison with the excluded**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Mather 2015		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(i) At least one known important domain was not appropriately measured, or not controlled for; **Models are adjusted for age, gender and education (SES) but also include self-rated health, which could be considered a mediator.**
Bias in selection of participants into the study	Serious	(ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation rate is 60% and no comparison is provided by authors between eligible population and participants.**
Bias in classification of interventions	Low	(i) Intervention status is well defined; *and* (ii) Intervention definition is based solely on information collected at the time of intervention. **Exposure was measured with a validated translation of Karasek's scale, participants on sick leave or on disability pension were excluded at baseline, and an adjustment was made for sick leave in the 2 years prior to baseline.**
Bias due to missing data	Serious	(i) Proportions of missing participants differ substantially across interventions: **64% of baseline participants included in the analyses**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Janssens 2014		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(i) At least one known important domain was not appropriately measured, or not controlled for; **Backward type analysis for age, gender and education in addition to other factors. The final model includes age, gender and alcohol consumption, the latter is considered a potentially intermediate factor.**
Bias in selection of participants into the study	Critical	(ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation at baseline was 30%. An analysis of non-respondents revealed no significant differences in age and sex. However, more lower grade employees were non-respondents according to the reference cited in the article. This supposes a possible selection bias**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **Exposure were based on Karasek's and Siegrist's models, validation found in the 2009 Clumeck study (Belstress I) (Moreau 2004).** (ii) Intervention definition is based solely on information collected at the time of intervention: **Prevalent cases were not excluded. Analyses were adjusted for baseline depressive symptoms, which partially controls for this bias.**
Bias due to missing data	Moderate	(i) Proportions of and reasons for missing participants differ slightly across intervention groups; **88% of baseline participants were included in the analyses and authors provided comparison between included and excluded participants showing that loss to follow-up was similar in terms of exposure as well as self-reported absence at baseline.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Critical**	

© 2020 American Medical Association. All rights reserved.

Juvani 2014		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious for both level	(i) At least one known important domain was not appropriately measured, or not controlled for; **Both exposure measures (work unit level and individual level) are considered problematic. Models with the ERI measure include potential intermediate variables, while separate components of the scale was not adjusted for SES.**
Bias in selection of participants into the study	Individual level: Serious Work-unit level: Moderate	Individual level: (ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation of 68% without comparison with non-participants** Work-unit level: (i) Selection into the study may have been related to intervention and outcome; and the proportion of participants for which this was the case was too low to induce important bias: **Participation of 100%**
Bias in classification of interventions	Serious for both level	(i) Intervention status is not well defined; **Not a validated proxy of the ERI questionnaire. Prevalent cases were not excluded, but long sick leaves (>90 days) were excluded and analyses were adjusted for baseline mental health.**
Bias due to missing data	Individual level: Serious Work-unit level: Moderate	Individual level: (i) Proportions of missing participants differ substantially across interventions: **58% of baseline participants included in the analyses** Work-unit level: (i) Proportions of and reasons for missing participants differ slightly across intervention groups: **86% of baseline participants included in the analyses**
Bias in measurement of outcomes	Low for both level	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Individual level: Serious** **Work-unit level: Serious**	

© 2020 American Medical Association. All rights reserved.

Ndjaboue 2014		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Adjusted for age, sex and education.**
Bias in selection of participants into the study	Moderate	(ii) A potentially important amount of follow-up time is missing from analyses: **Participation of 81% at baseline, without comparison with non-participants**
Bias in classification of interventions	Low	(i) Intervention status is well defined: **Exposure is defined with a validated French version of the ERI questionnaire.** *and* (ii) Intervention definition is based solely on information collected at the time of intervention**:** **Prevalent cases were excluded from the analyses**
Bias due to missing data	Moderate	(i) Proportions of and reasons for missing participants differ slightly across intervention groups; **81% of baseline participant included in the analyses, without comparison with excluded participants**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Moderate**	

© 2020 American Medical Association. All rights reserved.

Elovainio 2013		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Authors provide a model with appropriate measures and control for age, sex and SES (occupation) without potentially intermediate variables.**
Bias in selection of participants into the study	Serious	(ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation of 68% at baseline, without comparison with non-participants**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **Validated tool but without validation of the translation, prevalent cases excluded.**
Bias due to missing data	Serious	(i) Proportions of missing participants differ substantially across interventions; *or* Reasons for missingness differ substantially across interventions**: 55% of baseline participants included in the analyses, without comparison with excluded participants**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Samuelsson 2013		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **All important domains (age, gender and SES) are controlled for.**
Bias in selection of participants into the study	Moderate	(i) Selection into the study may have been related to intervention and outcome; *and* the proportion of participants for which this was the case was too low to induce important bias **Participation of 100% at baseline**
Bias in classification of interventions	Serious	(i) Intervention status is not well defined: **Exposure was assessed from a Matrix based on scores derived from another population, introducing misclassification.**
Bias due to missing data	Low	(i) Data were reasonably complete: **Data were 99% complete. 1% of data were missing on outcome.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Lahelma 2012		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Fully adjusted models control for age (as time scale), sex and occupation (SES).**
Bias in selection of participants into the study	Serious	(ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation of 67% at baseline, eligible population is different than participants. A larger portion of non-participants are younger employees, men, and more prone to sick leave.**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **Validated tool but without validation of the translation, prevalent cases were excluded.**
Bias due to missing data	Serious	(i) Proportions of missing participants differ substantially across interventions; *or* Reasons for missingness differ substantially across interventions**: Follow-up of 74%, with comparison that show slight differences but with no indication of bias. Missing data partially imputed**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Mantyniemi 2012		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate for both levels	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Models are adjusted for age, gender and SES**
Bias in selection of participants into the study	Individual level: Serious Work Level: Moderate	Individual level: (ii) A substantial amount of follow-up time is likely to be missing from analyses: **Participation of 68% with comparison that showed differences between participants and non-participants** Work Level: (i) Selection into the study may have been related to intervention and outcome; *and* the proportion of participants for which this was the case was too low to induce important bias: **Participation of 100%**
Bias in classification of interventions	Serious for both levels	(i) Intervention status is not well defined: **Proxy of validated questionnaire without validation provided and use of exposure matrix**
Bias due to missing data	Low for both levels	(i) Data were reasonably complete: **98% of baseline participants for matrix exposure and 94% of baseline participants for individual exposure were included in the analyses, with sensibility analyses that show no difference in the association reported between included and excluded participants.**
Bias in measurement of outcomes	Low for both levels	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Individual level: Serious** **Work level: Serious**	

© 2020 American Medical Association. All rights reserved.

Inoue 2010		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(i) At least one known important domain was not appropriately measured, or not controlled for: **Population included only male and appropriate adjustment provided for age and education (SES), but adjusted models included chronic physical condition, which can be considered as a potential mediator and may introduce residual confounding.**
Bias in selection of participants into the study	Moderate	A potentially important amount of follow-up time is missing from analyses: **Participation of 85% without comparison between participants and non-participants**.
Bias in classification of interventions	Low	(i) Intervention status is well defined: **Exposure was measured with a Japanese version of the NIOSH-GJSQ with validation provided.** *and* (ii) Intervention definition is based solely on information collected at the time of intervention: **Exclusion of prevalent cases and analyses are adjusted for depression at baseline**
Bias due to missing data	Moderate	(i) Proportions of and reasons for missing participants differ slightly across intervention groups: **81% of baseline participants included in the analyses with comparison that showed differences between included and excluded participants.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Kivimaki 2010		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Adjustment for gender and SES inherent to the study design (population included only female nurses) and model adjusted for age.**
Bias in selection of participants into the study	Serious	A potentially important amount of follow-up time is missing from analyses: **Participation rate is 78%. No comparison of participant and eligible population is provided by authors**
Bias in classification of interventions	Serious	(i) Intervention status is not well defined: **Exposure was measured with the short version of the psychological demands scale from Karasek's questionnaire, but no validated reference was provided**
Bias due to missing data	Low	(i) Data were reasonably complete: **95% at follow-up**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Clumeck 2009		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(i) At least one known important domain was not appropriately measured, or not controlled for; **Models are adjusted for age, living situations and occupation and are stratified by gender. Age was dichotomized at the median, resulting in potentially important residual confounding.**
Bias in selection of participants into the study	Critical	(ii) A substantial amount of follow-up time is likely to be missing from analyses**: Participation at baseline was 48% and no information is provided on refusals**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **JCQ version validated with supporting reference.** *and* (ii) Some aspects of the assignments of intervention status were determined retrospectively: **Prevalent cases were not excluded from analyses. Adjustment for mental health at baseline was performed with the CESD scale.**
Bias due to missing data	Moderate	(i) Proportions of and reasons for missing participants differ slightly across intervention groups: **Data were complete for 90% of the sample. Authors performed imputation to reduce missing data on exposure, but their analyses are unlikely to have removed all bias from other missing data. Little information is provided on missing data for covariates and no comparison is provided.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Critical**	

© 2020 American Medical Association. All rights reserved.

Wammstrom 2009		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Critical	(i) Confounding inherently not controllable: **Confounding not evaluated, only crude results presented**
Bias in selection of participants into the study	Serious	(ii) A substantial amount of follow-up time is likely to be missing from analyses**: Participation of 65% with comparison that shows differences between participants and non-participants**
Bias in classification of interventions	Low	(i) Intervention status is well defined: **This is a validation study**, *and* (ii) Intervention definition is based solely on information collected at the time of intervention: **Prevalent cases were excluded.**
Bias due to missing data	Moderate	(i) Proportions of and reasons for missing participants differ slightly across intervention groups: **Data are 85% complete after partial imputation**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Critical**	

© 2020 American Medical Association. All rights reserved.

Bourbonnais 2005		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Population is composed of 94% women and all are nurses. SES is therefore controlled for by the study design. It is unlikely that gender introduces important confounding. Models are adjusted for age.**
Bias in selection of participants into the study	Moderate	A potentially important amount of follow-up time is missing from analyses: **Participation rate at baseline was 70%, with a comparison showing that non-participants and participants are similar for work shift but not for job status, full time have a greater % of participation than part-time.**
Bias in classification of interventions	Moderate	((i) Intervention status is well defined; **Exposure was measured with the validated French version of Karasek's questionnaire as well as Siegrist's original questionnaire for reward component.** *and* (ii) Some aspects of the assignments of intervention status were determined retrospectively. **Prevalent cases were not excluded, but analyses partially adjust for prior mental health problems by controlling for previous all-cause work absence.**
Bias due to missing data	Serious	(i) Proportions of missing participants differ substantially across interventions; **Participation at follow-up was 66%** Reasons for missing differ substantially across interventions: **Authors provide a comparison between participants with complete data and participants with missing data that were excluded. The comparison shows that they are relatively similar in terms of job status exposure, job change, and psychological distress.** **Differences were noted in terms of age and seniority.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Boedeker 2001		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Adjusted analyses for the three important factors (age, sex and education as a proxy of SES) were appropriate and included in a model without potential intermediate factors. Data obtained by register.**
Bias in selection of participants into the study	Moderate	Selection into the study may have been related to intervention and outcome; *and* the proportion of participants for which this was the case was too low to induce important bias: **Participation rate is 100% due to the use of a national register**
Bias in classification of interventions	Serious	(i) Intervention status is not well defined; **Unclear validation of the measure to assess exposure or the job-exposure matrix.** (ii) Major aspects of the assignments of intervention status were determined in a way that could have been affected by knowledge of the outcome. **Prevalent cases were not excluded from analysis.**
Bias due to missing data	Low	(i) Data were reasonably complete: **Data obtained by register were complete. 100%.**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Bourbonnais 2001		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(i) At least one known important domain was not appropriately measured, or not controlled for: **Age was assessed but not included in the model. Models presented are crude, although there is some degree of control for SES and gender due to the study design.**
Bias in selection of participants into the study	Serious	A potentially important amount of follow-up time is missing from analysis: **Participation at baseline was 62%. Subjects who refused to participate differed from participants in terms of seniority, of employment status and types of work schedule. Also, a sensitivity analysis with hospitals with a better participation rate did not show any difference.**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **Exposure measured with a validated French version of Karasek's questionnaire.** and (ii) Some aspects of the assignments of intervention status were determined retrospectively: **Prevalent cases were not excluded and no sensitivity analyses were performed**
Bias due to missing data	Low	(i) Data were reasonably complete: **95% of baseline participants included in the analyses**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Stansfeld 1997		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Moderate	(i) Confounding expected, all known important confounding domains appropriately measured and controlled for: **Adjusted for age, gender and occupational grade (SES)**
Bias in selection of participants into the study	Serious	A potentially important amount of follow-up time is missing from analyse: **Participation at baseline of 73% without comparison with non-participants**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **partial validation of the tool used,** and (ii) Some aspects of the assignments of intervention status were determined retrospectively: **Prevalent were cases not excluded**
Bias due to missing data	Moderate	(i) Proportions of and reasons for missing participants differ slightly across intervention groups: **86% of baseline participants included in the analyses**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

Bourbonnais 1995		
Type of bias	**Classification**	**Reason - explanation**
Bias due to confounding	Serious	(ii) Reliability or validity of measurement of an important domain was low enough that we expect serious residual confounding: **Authors used the forward technique to control for potential confounding. Important domains of confounding were tested one at a time by stratification. This may lead to potential residual confounding.**
Bias in selection of participants into the study	Serious	A potentially important amount of follow-up time is missing from analyse: **Participation of 65% at baseline, comparison showed that participants and non-participants were not significantly different by age or sex.**
Bias in classification of interventions	Moderate	(i) Intervention status is well defined: **Authors provided references to support the validation of the French version of the questionnaire.** and (ii) Some aspects of the assignments of intervention status were determined retrospectively: **Prevalent cases were not excluded in the main analyses or in the secondary analyses**.
Bias due to missing data	Serious	(i) Proportions of missing participants differ substantially across interventions; *or* Reasons for missing differ substantially across interventions: **74% of baseline participants included in the analyses, with comparison that show no differences between participants and refusals**
Bias in measurement of outcomes	Low	(i) The methods of outcome assessment were comparable across intervention groups; *and* (ii) The outcome measure was unlikely to be influenced by knowledge of the intervention received by study participants (i.e. is objective) or the outcome assessors were unaware of the intervention received by study participants; *and* (iii) Any error in measuring the outcome is unrelated to intervention status. **Information obtained by registers, without knowledge of exposure.**
Overall	**Serious**	

© 2020 American Medical Association. All rights reserved.

eTable 2. Results of the Sensitivity Analyses

Analysis	Exposure	Studies included	N, Meta-estimates, heterogeneity (I^2)		
			Absence	Disability pension	Total
Same as main analyses but stratified for absence or disability pension	Job strain	Same as main analyses	n=7 1.41 [1.01, 1.97] I^2 = 64%	n=1 1461 [1.25, 1.71] I^2 = NA	n=9 1.47 [1.24, 1.74] I^2 = 58%
	High psychological demands	Same as main analyses	n=5 1.24 [1.03, 1.49] I^2 = 62%	n=2 1.15 [0.65, 2.01] I^2 = 65%	n=7 1.03 [0.79, 1.33] I^2 = 56%
	Low job control	Same as main analyses	n=4 1.17 [0.96, 1.43] I^2 = 70%	n=1 1.67 [1.12, 2.49] I^2 = NA	n=5 1.25 [1.02, 1.53] I^2 = 72%
	Low social support at work	Same as main analyses	n=8 1.14 [1.02, 1.28] I^2 = 42%	n=1 0.82 [0.56, 1.20] I^2 = NA	N=9 1.12 [0.99, 1.26] I^2 = 45%
	Effort-reward imbalance	Same as main analyses	N=2 1.50 [1.24, 1.83] I^2 = 0%	N=1 1.90 [1.51, 2.39] I^2 = NA	N=3 1.66 [1.37, 2.00] I^2 = 34%
	Low reward	Same as main analyses	N=2 1.64 [1.32, 2.06] I^2 = 0%	N=1 1.92 [1.49, 2.47] I^2 = NA	N=3 1.76 [1.49, 2.08] I^2 = 0%
Including studies that evaluated aggregated level of exposure	Job strain	Same as main analysis + Samuelsson (2013)	n=7 1.41 [1.01, 1.97] I^2 = 64%	n=2 1.31 [1.04, 1.66] I^2 = 82%	n=9 1.38 [1.15, 1.65] I^2 = 71%
	High psychological demands	Same as main analysis + Samuelsson (2013) and Boedeker (2001)	n=6 1.03 [0.67, 1.56] I^2 = 95%	n=3 1.03 [0.74, 1.43] I^2 = 67%	n=9 1.03 [0.79, 1.33] I^2 = 93%
	Low job control	Same as main analysis + Samuelsson (2013) and Boedeker (2001)	n=5 1.60 [1.08, 2.37] I^2 = 94%	n=2 1.29 [0.86, 1.92] I^2 = 76%	n=7 1.45 [1.16, 1.81] I^2 = 92%
	Low social support at work	Same as main analysis + Samuelsson (2013)	n=8 1.14 [1.02, 1.28] I^2 = 42%	n=2 0.89 [0.81, 0.98] I^2 = 0%	N=10 1.07 [0.95, 1.22] I^2 = 65%

© 2020 American Medical Association. All rights reserved.

Analysis	Exposure	Studies included	N, Meta-estimates, heterogeneity (I^2)		
			Absence	Disability pension	Total
Including only studies using hazard ratios	Job strain	Bourbonnais 2005, Inoue, Ndjaboue 2017, Mantyniemi	n=3 1.83 [1.08, 3.08] I^2 = 65%	n=1 1.46 [1.25, 1.71] I^2 = NA	n=4 1.61 [1.31, 1.97] I^2 = 72%
Including studies using quadrant categorization	Job strain	Bourbonnais (1995, 2001 and 2005), Mather, Ndjaboue 2017	n=5 1.49 [1.07, 2.08] I^2 = 62%	NA	n=5 1.49 [1.07, 2.08] I^2 = 62%
Without studies on effort or without other tools than JCQ	High psychological demands	Same as main analysis, without Ndjaboue 2014, Juvani 2014, and Van Hoffen	n=3 1.41 [1.18, 1.67] I^2 = 18%	n=1 1.48 [1.00, 2.19] I^2 = NA	n=4 1.39 [1.25, 1.54] I^2 = 0%
With results on co-workers' support instead of supervisors' (Inoue, van Hoffen)	Low social support at work	Same as main analysis	n=8 1.13 [1.00, 1.28] I^2 = 46%	n=1 0.82 [0.56, 1.20] I^2 = NA	n=9 1.11 [0.95, 1.25] I^2 = 48%
Including studies with a critical risk of bias : Roelen 2018, Vendrig 2018, Janssens 2014, Clumeck 2009, and Wannstroms 2009	Job strain	Same as main analysis + , Janssens 2014, Clumeck 2009,	N=9 1.50 [1.13, 1.99] I^2 = 71%	N=1 1.46 [1.25;1.71] I^2 = NA	N=10 1.48 [1.25,1.71] I^2 = 67%
	High psychological demands	Same as main analysis + : Roelen 2018, Vendrig 2018, Janssens 2014, Clumeck 2009	N=9 1.15 [1.04,1.27] I^2 = 63%	N=2 1.15 [0.65,2.01] I^2 = 65%	N=11 1.15 [1.04, 2.01] I^2 = 60%
	Low job control	Same as main analysis + Vendrig 2018, Janssens 2014, Clumeck 2009	N=7 1.23 [1.07, 1.42] I^2 = 69%	N=1 1.67 [1.12, 2.49] I^2 = NA	N=8 1.27 [1.10, 1.46] I^2 = 70%
	Low social support at work	Same as main analysis + Roelen 2018, Vendrig 2018, Janssens 2014, Clumeck 2009	N=12 1.10 [1.04, 1.17] I^2 = 25%	N=1 0.82 [0.56, 1.20] I^2 = NA	N=13 1.09 [1.03, 1.16] I^2 = 28%
	Effort-reward imbalance	Same as main analysis + Janssens 2014	N=3 1.35 [1.07, 1.71] I^2 = 59%	N=1 1.90 [1.51, 2.39] I^2 = NA	N=4 1.49 [1.16, 1.91] I^2 = 74%
	Low reward	Same as main analysis + Janssens 2014	N=3 1.48 [1.24, 1.76] I^2 = 12%	N=1 1.92 [1.49, 2.47] I^2 = NA	N=4 1.61 [1.34, 1.93] I^2 = 42%

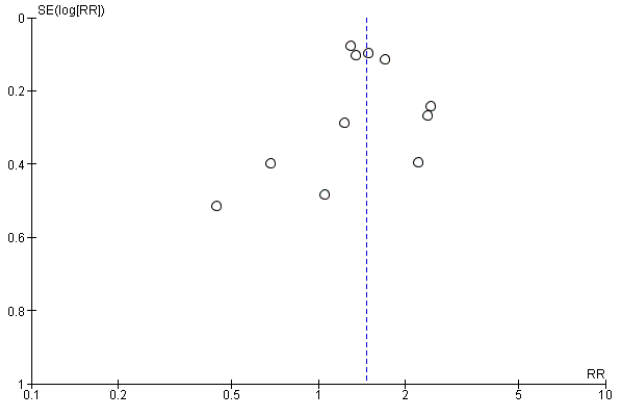
© 2020 American Medical Association. All rights reserved.

Analysis	Exposure	Studies included	N, Meta-estimates, heterogeneity (I^2)		
			Absence	Disability pension	Total
Excluding studies that did not clearly state that prevalent cases were excluded	Job strain	Without Bourbonnais 1995, 2001 and 2005, and Otha 2017	N=3 1.54 [0.98, 2.44] I^2 = 70%	N=1 1.46 [1.25, 1.71] I^2 = NA	N=4 1.49 [1.27, 1.75] I^2 = 58%
	High psychological demands	Without Bourbonnais 2005	N=4 1.17 [0.97, 1.41] I^2 = 59%	N=2 1.15 [0.65, 2.01] I^2 = 65%	N=6 1.17 [0.99, 1.38] I^2 = 51%
	Low job control	Without Bourbonnais 2005	N=3 1.13 [0.90, 1.43] I^2 = 75%	N=1 1.67 [1.12, 2.49] I^2 = NA	N=4 1.24 [0.98, 1.57]
	Low social support at work	Without Bourbonnais 1995, 2001 and 2005, Otha 2017, and Stansfeld 1997	N=3 1.10 [0.99, 1.21] I^2 = 0%	N=1 0.82 [0.56, 1.20] I^2 = NA	N=4 1.08 [0.98, 1.19] I^2 = 0%
	Effort-reward imbalance	Without Bourbonnais 2005	NA, N=1	NA, N=1	N=2 1.64 [1.23, 2.19] I^2 = 67%
	Low reward	Without Bourbonnais 2005	NA, N=1	NA, N=1	N=2 1.75 [1.43, 2.14] I^2 = 12%

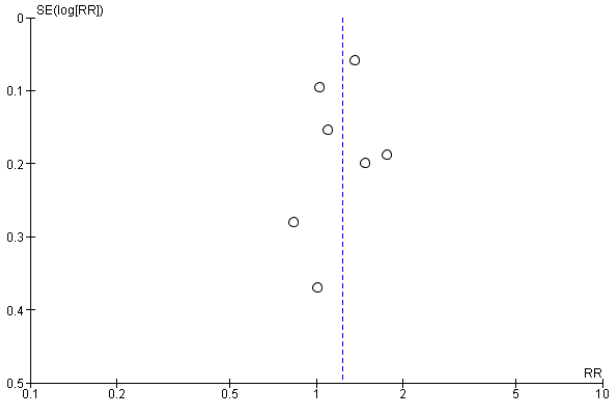
© 2020 American Medical Association. All rights reserved.

eFigure 2. Funnel Plot of Studies on the Effect of A) Job Strain, B) High Psychological Demands and C) Low Social Support at Work on the Risk of Sickness Absence Due to a Diagnosed Mental Disorder

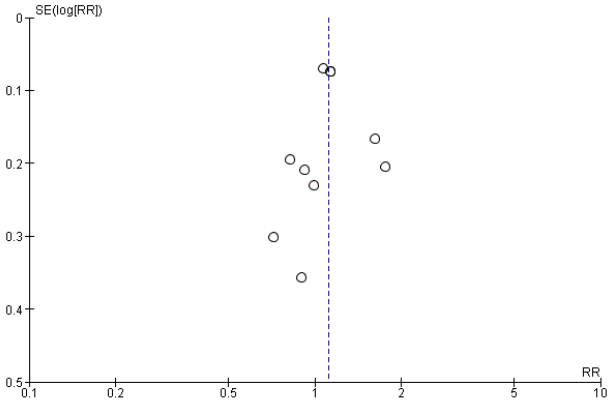
A)



B)



C)



© 2020 American Medical Association. All rights reserved.

eReferences

1. Roelen CAM, van Hoffen MFA, Waage S, et al. Psychosocial work environment and mental health-related long-term sickness absence among nurses. *Int Arch Occup Environ Health.* 2018;91(2):195-203.
2. Brisson C, Blanchette C, Guimont C, Dion J, Moisan J, Vézina M. Reliability and Validity of the French Version of the 18-item Karasek Job Content Questionnaire. *Work Stress.* 1998;12(4):322-336.
3. Vendrig AA, Schaafsma FG. Reliability and Validity of the Work and Well-Being Inventory (WBI) for Employees. *Journal of occupational rehabilitation.* 2018;28(2):377-390.
4. van Hoffen MFA, Roelen CAM, van Rhenen W, Schaufeli WB, Heymans MW, Twisk JWR. Psychosocial work characteristics and long-term sickness absence due to mental disorders. *Journal of mental health (Abingdon, England).* 2018:1-8.
5. Veldhoven MV, Jonge JD, Broersen S, Kompier M, Meijman T. Specific relationships between psychosocial job conditions and job-related stress: A three-level analytic approach. *Work & Stress: An International Journal of Work, Health & Organisations.* 2002;16(3):207-228.
6. Ndjaboue R, Brisson C, Talbot D, Vezina M. Combined exposure to adverse psychosocial work factors and medically certified absence for mental health problems: A 5-year prospective study. *J Psychosom Res.* 2017;92:9-15.
7. Karasek R, Brisson C, Kawakami N, Houtman I, Bongers P, Amick B. The Job Content Questionnaire (JCQ): an instrument for internationally comparative assessments of psychosocial job characteristics. *J Occup Health Psychol.* 1998;3(4):322-355.
8. Niedhammer I, Siegrist J, Landre MF, Goldberg M, Leclerc A. [Psychometric properties of the French version of the Effort-Reward Imbalance model]. *Rev Epidemiol Sante Publique.* 2000;48(5):419-437.
9. Ohta M, Higuchi Y, Kumashiro M, Yamato H, Sugimura H. Decrease in Work Ability Index and sickness absence during the following year: a two-year follow-up study. *Int Arch Occup Environ Health.* 2017;90(8):883-894.
10. Kawakami N, Takao S, Kobayashi Y, Tsutsumi A. Effects of web-based supervisor training on job stressors and psychological distress among workers: a workplace-based randomized controlled trial. *Journal of occupational health.* 2006;48(1):28-34.
11. Urakawa K, Yokoyama K, Itoh H. Sense of coherence is associated with reduced psychological responses to job stressors among Japanese factory workers. *BMC Res Notes.* 2012;5:247.
12. Juvani A, Oksanen T, Virtanen M, et al. Organizational justice and disability pension from all-causes, depression and musculoskeletal diseases: A Finnish cohort study of public sector employees. *Scand J Work Environ Health.* 2016;42(5):395-404.
13. Mather L, Bergstrom G, Blom V, Svedberg P. High Job Demands, Job Strain, and Iso-Strain are Risk Factors for Sick Leave due to Mental Disorders: A Prospective Swedish Twin Study With a 5-Year Follow-Up. *J Occup Environ Med.* 2015;57(8):858-865.
14. Sanne B, Torp S, Mykletun A, Dahl AA. The Swedish Demand-Control-Support Questionnaire (DCSQ): factor structure, item analyses, and internal consistency in a large population. *Scandinavian journal of public health.* 2005;33(3):166-174.
15. Janssens H, Clays E, De Clercq B, et al. The relation between psychosocial risk factors and cause-specific long-term sickness absence. *Eur J Public Health.* 2014;24(3):428-433.
16. Moreau M, Valente F, Mak R, et al. Occupational stress and incidence of sick leave in the Belgian workforce: the Belstress study. *J Epidemiol Community Health.* 2004;58(6):507-516.
17. Juvani A, Oksanen T, Salo P, et al. Effort-reward imbalance as a risk factor for disability pension: the Finnish Public Sector Study. *Scand J Work Environ Health.* 2014;40(3):266-277.

© 2020 American Medical Association. All rights reserved.

18. Ndjaboue R, Brisson C, Vezina M, Blanchette C, Bourbonnais R. Effort-reward imbalance and medically certified absence for mental health problems: a prospective study of white-collar workers. *Occup Environ Med.* 2014;71(1):40-47.
19. Elovainio M, Linna A, Virtanen M, et al. Perceived organizational justice as a predictor of long-term sickness absence due to diagnosed mental disorders: results from the prospective longitudinal Finnish Public Sector Study. *Soc Sci Med.* 2013;91:39-47.
20. Samuelsson A, Ropponen A, Alexanderson K, Svedberg P. Psychosocial working conditions, occupational groups, and risk of disability pension due to mental diagnoses: a cohort study of 43,000 Swedish twins. *Scand J Work Environ Health.* 2013;39(4):351-360.
21. Johnson JV, Stewart WF. Measuring work organization exposure over the life course with a job-exposure matrix. *Scand J Work Environ Health.* 1993;19(1):21-28.
22. Lahelma E, Laaksonen M, Lallukka T, et al. Working conditions as risk factors for disability retirement: a longitudinal register linkage study. *BMC Public Health.* 2012;12:309.
23. Mantyniemi A, Oksanen T, Salo P, et al. Job strain and the risk of disability pension due to musculoskeletal disorders, depression or coronary heart disease: a prospective cohort study of 69,842 employees. *Occup Environ Med.* 2012;69(8):574-581.
24. Inoue A, Kawakami N, Haratani T, et al. Job stressors and long-term sick leave due to depressive disorders among Japanese male employees: findings from the Japan Work Stress and Health Cohort study. *J Epidemiol Community Health.* 2010;64(3):229-235.
25. Haratani T, Kawakami N, Araki S, et al. Psychometric properties and stability of the Japanese version of the NIOSH job stress questionnaire. 25th International Congress on Occupational Health - Book of Abstracts; 1996.
26. Kivimaki M, Vahtera J, Kawachi I, et al. Psychosocial work environment as a risk factor for absence with a psychiatric diagnosis: an instrumental-variables analysis. *Am J Epidemiol.* 2010;172(2):167-172.
27. Clumeck N, Kempenaers C, Godin I, et al. Working conditions predict incidence of long-term spells of sick leave due to depression: results from the Belstress I prospective study. *J Epidemiol Community Health.* 2009;63(4):286-292.
28. Wannstrom I, Peterson U, Asberg M, Nygren A, Gustavsson JP. Psychometric properties of scales in the General Nordic Questionnaire for Psychological and Social Factors at Work (QPS): confirmatory factor analysis and prediction of certified long-term sickness absence. *Scand J Psychol.* 2009;50(3):231-244.
29. Bourbonnais R, Brisson C, Vézina M, Masse B, Blanchette C. Psychosocial Work Environment and Certified Sick Leave among Nurses during Organizational Changes and Downsizing. *Relations Industrielles / Industrial Relations.* 2005;60(3):483-509.
30. Bourbonnais R, Vinet A, Vezina M, Gingras S. Certified sick leave as a non-specific morbidity indicator: a case-referent study among nurses. *Br J Ind Med.* 1992;49(10):673-678.
31. Boedeker W. Associations between workload and diseases rarely occurring in sickness absence data. *J Occup Environ Med.* 2001;43(12):1081-1088.
32. Bodeker W, Friedel H, Friedrichs M, Rottger C. *Kosten der Frühberentung.* Dortmund/Berlin: Wirtschaftsverlag.; 2006.
33. Bourbonnais R, Mondor M. Job strain and sickness absence among nurses in the province of Quebec. *Am J Ind Med.* 2001;39(2):194-202.
34. Stansfeld SA, Rael EG, Head J, Shipley M, Marmot M. Social support and psychiatric sickness absence: a prospective study of British civil servants. *Psychol Med.* 1997;27(1):35-48.
35. Marmot MG, Smith GD, Stansfeld S, et al. Health inequalities among British civil servants: the Whitehall II study. *Lancet.* 1991;337(8754):1387-1393.
36. Bourbonnais R, Vezina M. [Mental health of white collar workers and the psychosocial environment at work]. *Sante Ment Que.* 1995;20(2):163-184.

© 2020 American Medical Association. All rights reserved.